

Geodesic MDS followed by edge-KK

Radius, neighborhood, sampling, and scale sensitivity

Source: `~/current_projects/grip/tools/experiments/mds-edge-kk-radius/report.tex`

Build 2026-09-05 16:33:23 EDT

Question and findings

Does refining an MDS embedding with local edge constraints restore the original paraboloid or saddle? How does that answer depend on the neighborhood size, the distance used to build the graph, and the optimizer's scale policy? We examine $(x, y, x^2 + y^2)$ and $(x, y, x^2 - y^2)$ on disks of increasing radius. Smooth-surface geodesics define one graph regime; ambient Euclidean chords define the other. All fitted configurations have three coordinates.

The graph regime materially changes the geometric answer. At $r = 64$, $k = 32$, the fixed-scale stress-MDS-to-edge-KK branch gives the following base-disk medians across independent samples. Coordinate error uses similarity alignment; path/reference error compares edge-calibrated retained paths with smooth geodesics.

Surface	Graph regime	s_2/s_1	s_3/s_2	Coordinate (%)	Path/reference (%)
Paraboloid	geodesic	0.038	0.846	1.46	0.34
Paraboloid	ambient	0.028	0.915	0.88	0.38
Saddle	geodesic	0.901	0.896	80.38	0.36
Saddle	ambient	0.019	0.917	0.90	37.00

Across 1176 primary $n = 240$ graphs, profiled-scale edge-KK lowers fixed-path error in 1016 stress-initialized comparisons and increases it in 76; fixed-scale edge-KK lowers it in 1075 and increases it in 17. Changes smaller than 10^{-8} in absolute relative-error units are counted as ties. Good path fidelity is therefore neither a universal refinement improvement nor evidence of surface recovery.

The pilot also exposed a scale issue: the profiled, unnormalized edge objective can decrease by uniform contraction. Fixed-scale controls prevent this degeneracy and are essential when interpreting absolute loss, stopping values, and spatial extent.

These are paired finite-sample experiments. Increasing radius at fixed sample size does not maintain a resolved surface mesh. Neither a thin plot nor a small edge objective establishes a limiting dimension or recovery theorem.

Experimental design

We use radii $r = 1, 2, 4, 8, 16, 32, 64$, neighborhood sizes $k = 4, 8, 16, 32, 64, 128, 239$, and three independent samples of $n = 240$ points. The random seeds are 20260905–20260907. Within a sample, the radial quantiles and angles are coupled across radii and surfaces. The first sample reuses the audited original numerical geodesics. A nested $n = 480$ sample at $r = 64$ adds 240 independent points, retaining the original indices, and uses $k = 8, 16, 64, 128, 479$. Thus 8 and 64 test fixed k ; 16 and 128 match the original k/n at 8 and 64. Complete graphs are explanatory controls. This single nested size comparison does not estimate a large- n limit.

Both sampling measures are included. For uniform $U \in (0, 1)$ and independent uniform angle, base-disk sampling uses $\rho = r\sqrt{U}$. Surface-area sampling uses

$$\rho = \frac{1}{2} \left(\{1 + U[(1 + 4r^2)^{3/2} - 1]\}^{2/3} - 1 \right)^{1/2}.$$

Both graph surfaces have area element $\sqrt{1 + 4\rho^2} \rho d\rho d\theta$, which gives this inverse radial distribution by integration.

An undirected edge is included if either endpoint selects the other among its k nearest neighbors; ties are broken by vertex index. In the geodesic regime, both ranking and edge targets use the smooth-geodesic matrix Δ . In the ambient regime, both use original-coordinate chords. Disconnected unions are explicitly augmented by a minimum spanning tree under the same distances. No vertices are removed. Connectivity, augmentation, degree, edge lengths, and parameter-plane radial/angular separations are recorded.

Classical MDS and stress MDS use strict graph shortest distances D_G . A third initializer in the geodesic regime is classical scaling of full Δ , held fixed across k . Stress MDS uses the grip adapter to `smacof::mds(type="ratio")`: stress majorization with ratio disparities and internal normalization [Mair et al., 2022]. The adapter restores the raw-stress-optimal coordinate scale. Each graph has one classical and two Gaussian starts, at most 1,000 iterations per start and tolerance 10^{-8} . Gaussian draws are matched across k ; the adapter optimally rescales each start against its own target matrix. The least achieved raw stress selects the fit; global optimality is not claimed.

The pilot changed the scale controls

Every target and initializer is first divided by the RMS full smooth-geodesic distance. The declared edge-KK workflow uses density-to-uniform stiffness (0, .25, .5, .75, 1) with 200 steps per stage and a profiled target scale. A uniform-only run has the same total allowance of 1,000 steps. The original 72-graph pilot exposed severe contraction in some geodesic fits. We retained those outputs and repeated the pilot with additional fixed-target-scale runs, using both schedules. Those controls were then applied throughout the study. The package defaults were not changed.

Write $\ell_e > 0$ for edge targets, $d_e(Z) = \|z_i - z_j\|$ for embedded edge lengths, and $\kappa_e \geq 0$ for the current stiffness (nonnegative, with a positive weighted target norm). The implemented objective is

$$E(Z, a) = \frac{1}{2} \sum_e \kappa_e [d_e(Z) - a \ell_e]^2, \quad a(Z) = \frac{\sum_e \kappa_e d_e(Z) \ell_e}{\sum_e \kappa_e \ell_e^2}.$$

Since $d_e(cZ) = c d_e(Z)$, for $c > 0$ we have $a(cZ) = c a(Z)$ and

$$E(cZ, a(cZ)) = c^2 E(Z, a(Z)).$$

Thus uniform contraction reduces any positive profiled loss without changing shape or relative edge errors. A vanishing absolute objective or gradient is not a certificate of shape convergence. This statement follows directly from the implemented loss and is independently checked on saved pilot fits. It does not invalidate relative scores computed at a common calibrated scale. A scale-invariant alternative would divide the minimized numerator by $\sum_e \kappa_e d_e^2$, giving

$$Q(Z) = 1 - \frac{(\sum_e \kappa_e d_e \ell_e)^2}{(\sum_e \kappa_e d_e^2)(\sum_e \kappa_e \ell_e^2)}.$$

When $\sum_e \kappa_e d_e^2 > 0$, this also equals raw edge stress minimized over a coordinate multiplier, divided by the weighted target norm squared. The identity follows by minimizing a scalar quadratic. It explains why fixing target scale is a valid control; it does not make unnormalized profiled loss scale invariant or equate the finite-budget trajectories.

Fixed-scale controls use $a = 1$, so contraction cannot remove a positive target. For a complete graph with uniform stiffness, $2E(Z, 1)$ equals all-pair raw distance stress when the same targets are used. Tiny numerical violations of triangle inequalities in smooth inputs can distinguish direct targets from strict graph distances; these discrepancies are measured separately. For complete ambient graphs, the original coordinates exactly realize the targets.

For shape displays and end-to-end path scoring, let $b = \sum_e d_e \ell_e / \sum_e \ell_e^2$. We divide fitted coordinates or path lengths by this edge-derived b , then restore physical units. No smooth-reference scale is fitted. Saved coordinates also retain the optimizer's returned scale. Plots use equal spatial units and at most one additional uniform r^2 divisor.

What the diagnostics mean

For observed length vector y and positive target vector t , define $a = y^\top t / (t^\top t)$. Edge and fixed-path relative errors are $\|y - at\|/\|at\|$, each with its own diagnostic scale. Every graph uses all unordered vertex pairs and the same retained routes for every candidate. The graph constructor's near-tie route convention can produce slightly longer routes than strict D_G ; MDS receives symmetric strict distances and route scoring uses the retained route lengths. These roles are not interchanged.

Chord raw stress is $\sum_{i < j} (d_{ij} - D_{G,ij})^2$ at returned physical scale. Profiled chord Stress-1 is $\|d - aD_G\|/\|d\|$. These are different statistics; raw stress after a contracting profiled refinement is not its optimized criterion. Graph/reference error is $\|D_G - \Delta\|/\|\Delta\|$ in physical units. End-to-end path/reference error is $\|L_Z/b - \Delta\|/\|\Delta\|$, where L_Z sums embedded edge lengths along retained routes. Small graph-to-layout error cannot substitute for graph-to-surface validation.

Coordinate error is relative Frobenius error after similarity alignment to known generating coordinates with known point correspondence. It is not a closest-surface distance or a test of topology. Let $s_1 \geq s_2 \geq s_3$ be singular values of the centered coordinates. A line limit requires $s_2/s_1 \rightarrow 0$; a nondegenerate plane requires $s_3/s_2 \rightarrow 0$ while s_2/s_1 stays nonzero. Both ratios are needed. We report finite-radius ratios and all three dimensional singular values.

A useful deterministic bound explains one implication of edge agreement. If every retained edge satisfies $|d_e/(b\ell_e) - 1| \leq \epsilon$, then every retained path γ satisfies

$$\left| \frac{L_Z(\gamma)}{b} - L_G(\gamma) \right| \leq \sum_{e \in \gamma} |d_e/b - \ell_e| \leq \epsilon \sum_{e \in \gamma} \ell_e = \epsilon L_G(\gamma).$$

This is a graph-to-layout statement. Small mean edge stress does not establish its maximum-relative-error hypothesis, and the bound does not identify a unique surface realization. A one-dimensional intrinsic metric can also require a higher-dimensional Euclidean approximation: intrinsic dimension and Euclidean span answer different questions. For example, four leaves at distance $t > 0$ from the center of a metric star have all interleaf distances $2t$. Their classical Gram matrix is $2t^2 J$, where $J = I - \mathbf{1}\mathbf{1}^\top/4$ has rank three. Thus even this four-point subset of an intrinsically one-dimensional tree requires three Euclidean coordinates for exact distance realization. The existing direct-geodesic MDS line/tree analyses therefore do not automatically determine edge-KK's limit.

Numerical reliability and optimizer sensitivity

The experiment contains 1216 primary graphs, 16416 scored primary candidates, and 3648 MDS starts. 74 graphs require MST augmentation; the largest retained-route/strict-distance discrepancy is $6.67\text{e-}08$ in normalized units. Independent edge/path calculations agree with grip to $1.25\text{e-}14$; explicit route sums agree to $8.88\text{e-}16$.

All 88 numerical input matrices pass symmetry, all-pair chord lower bounds, explicit surface-path upper bounds, and exhaustive sampled-vertex triangle checks at tolerance 10^{-7} times RMS distance. The largest normalized triangle violation is $8.49\text{e-}09$. Independent high-radius validation covers 256 actual pairs in 16 cases, including short/far and random pairs. The largest sampled relative discrepancy is $5.16\text{e-}08$. Saddle checks use tighter shooting and independent SciPy collocation/integration; paraboloid checks use independent quadrature. This sampled maximum is not an all-pair accuracy bound. A separate reconstruction of graph distances and diagnostics from 16928 saved coordinate arrays agrees to $2.59\text{e-}12$ after scaling dimensional discrepancies. Primary profiled fits reach edge-calibration factors as small as $4.07\text{e-}10$, whereas the fixed-scale controls range from 0.99339 to 1.02624. The contraction occurs at intermediate radii too, so it is not an asymptotic property of the surface.

At $r = 64$, replicate 1, $k = 8, 32, 128, 239$, additional fits start from tiny three-dimensional perturbations, independent random coordinates, and original coordinates. Perturbations have entrywise RMS 10^{-4} in normalized units. Separate continuations add 2,000 uniform-stiffness steps to each selected classical- and stress-initialized edge-KK result. Both scale policies are included. Random and original-coordinate configurations are shared across the classical and stress branch comparisons; their duplicate table rows are not independent replications. These exploratory controls do not replace primary fits. A planar start can have zero transverse gradient for a differentiable distance-based objective; a perturbation check addresses particular fits, not every planar global minimizer.

25/3648 MDS starts reach the iteration cap; the largest between-start difference in target-normalized RMSE is 1.530 percentage points. Iteration limits and small stress changes do not certify global optimality. The additional controls contain 512 scored candidates on 32 graphs. Under profiled scale, the largest s_3/s_2 change is 0.0003 for tiny perturbations, 0.0124 for extra steps, and 0.7807 for independent random starts. Under fixed scale, the largest s_3/s_2 change is 0.0003 for tiny perturbations, 0.0263 for extra steps, and 0.8208 for independent random starts. These distinguish local stability around MDS starts from dependence on the choice of a different starting configuration.

Neighborhood, sampling, and sample size

At $r = 64$, the fixed-scale stress-initialized branch gives the following ranges across all seven k values, both sampling measures, and three samples. Paraboloid, geodesic graphs: s_2/s_1 ranges from 0.035 to 0.100, and s_3/s_2 from 0.507 to 0.960. Paraboloid, ambient graphs: s_2/s_1 ranges from 0.027 to 0.089, and s_3/s_2 from 0.595 to 0.958. Saddle, geodesic graphs: s_2/s_1 ranges from 0.839 to 0.956, and s_3/s_2 from 0.835 to 0.982. Saddle, ambient graphs: s_2/s_1 ranges from 0.018 to 0.069, and s_3/s_2 from 0.604 to 0.984. These are observed finite-radius ranges, not limiting-dimension estimates.

Holding the full-geodesic classical initializer fixed across k still gives a nonzero path/reference-error range (above 10^{-8}) in 84/84 cases. Thus neighborhood effects are not explained solely by changing the MDS input. The initializer-control figures compare this branch with graph-dependent classical starts.

Increasing sample size does not hold graph locality fixed. In the four nested geodesic cases below, keeping $k = 8$ increases path/reference error, whereas doubling k to preserve k/n reduces it relative to the $n = 240, k = 8$ baseline. Each row is one nested sample pair under fixed-scale stress-MDS-to-edge-KK; values are percentages, not medians across three samples.

Surface	Sampling	$n = 240, k = 8$	$n = 480, k = 8$	$n = 480, k = 16$
Paraboloid	Base disk	7.77	9.23	2.34
Paraboloid	Surface area	6.26	9.35	1.17
Saddle	Base disk	1.07	8.41	1.03
Saddle	Surface area	1.83	8.30	1.41

The higher- k and ambient comparisons are retained in the size figures and full table. One nested size increase does not establish a general sample-size law.

Stress initialization is not uniformly superior after refinement. With profiled scale, stress initialization gives lower final path error in 610/1176 comparisons, classical initialization in 404, with 162 ties at the declared tolerance. With fixed scale, stress initialization gives lower final path error in 590/1176 comparisons, classical initialization in 278, with 308 ties at the declared tolerance. The stratified table retains reversals by surface, sampling measure, and graph regime; these paired counts are not independent statistical trials.

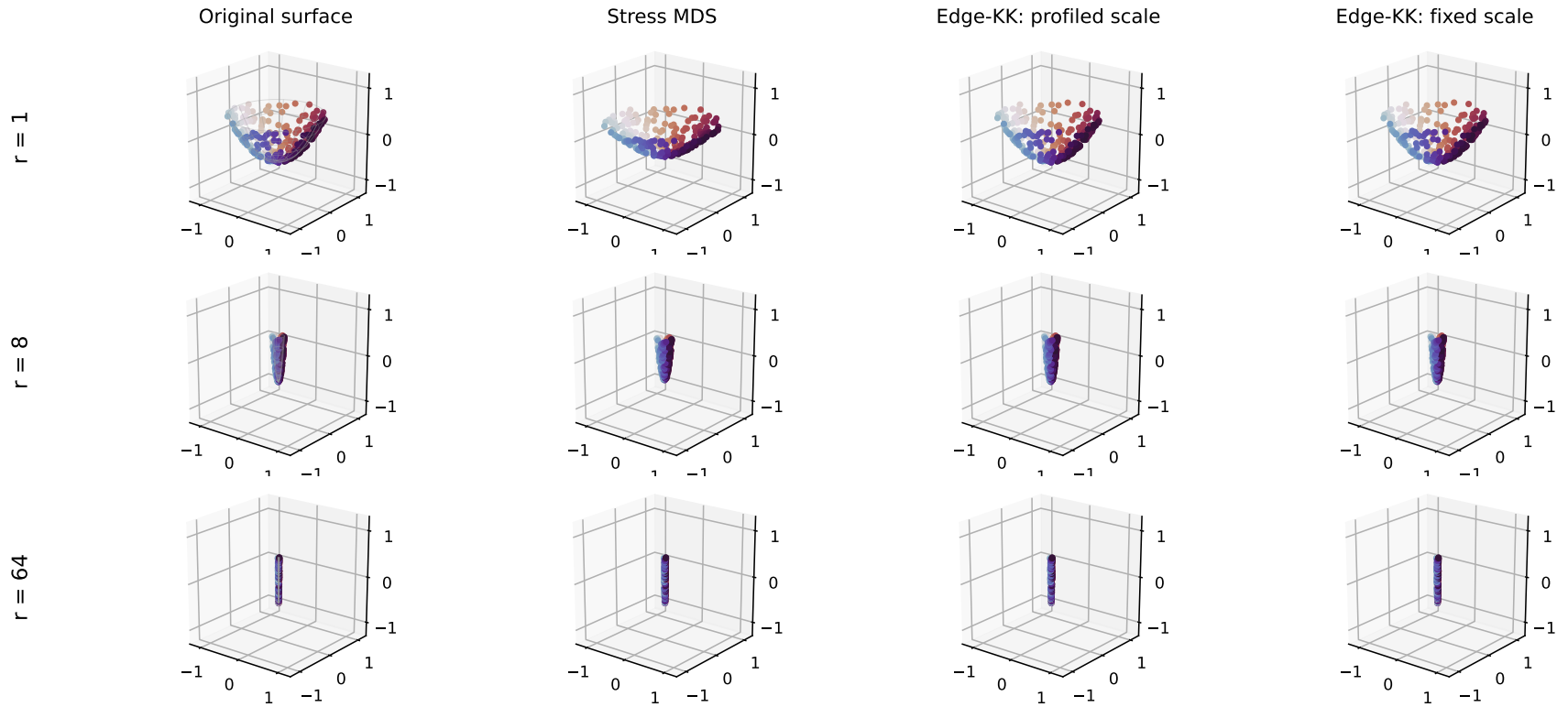
Implications for the grip software article

The results support presenting edge-KK as a graph-edge refinement with separately assessed path and geometric fidelity. They do not support a general claim that it reverses MDS flattening or recovers the generating manifold. The graph/reference discrepancy, neighborhood locality, and scale policy must be visible alongside the final graph-to-layout score. The complete ambient control is exact Euclidean realization, not evidence of intrinsic-geodesic recovery.

The full radius, k , sampling, and size study belongs in a supplement or a separate research report. A concise main-text comparison should distinguish classical scaling, stress MDS, and edge-only refinement; explain the graph regime; and avoid equating path fidelity with generating-surface recovery. The scale issue needs explicit treatment before using loss or stopping values as evidence for the profiled workflow. This report records an experiment and its controls; integration into the main manuscript is a later phase.

Paraboloid: geodesic graph, $k = 32$

Paraboloid · geodesic graph · $k=32$ · first base-disk sample
Edge-calibrated physical coordinates / r^2 ; equal spatial units and common limits

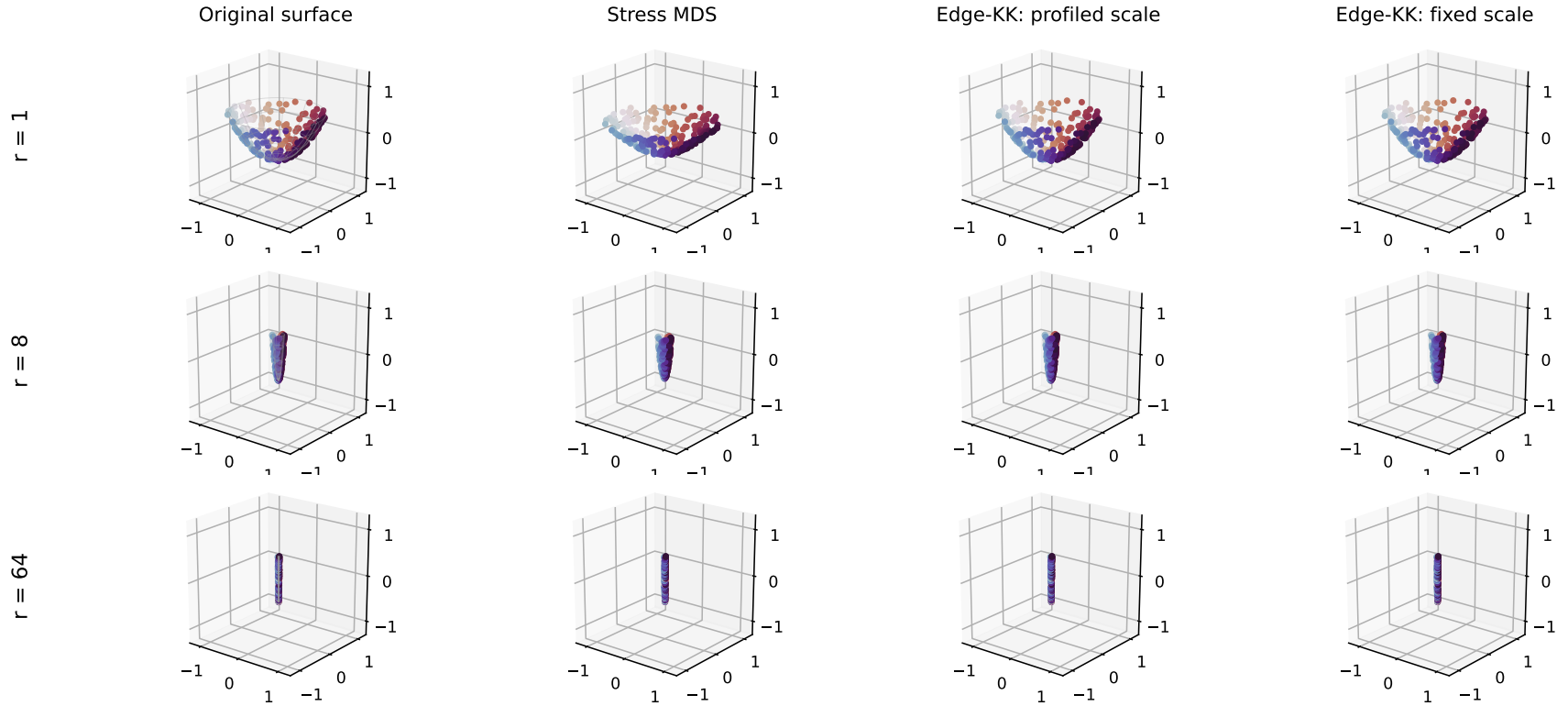


Original wireframe covers the full parameter disk. Thin equal-unit views do not mean the surface domain was truncated.

Sample 1, uniform base-disk sampling, $n=240$. Coordinates are aligned by an orthogonal transformation and translation, edge-calibrated, and divided uniformly by r^2 . All axes use equal units and common limits. The original wireframe spans the full disk. A thin equal-unit view does not truncate the original parameter domain. Colors preserve vertex correspondence.

Paraboloid: ambient graph, $k = 32$

Paraboloid · ambient graph · $k=32$ · first base-disk sample
Edge-calibrated physical coordinates / r^2 ; equal spatial units and common limits

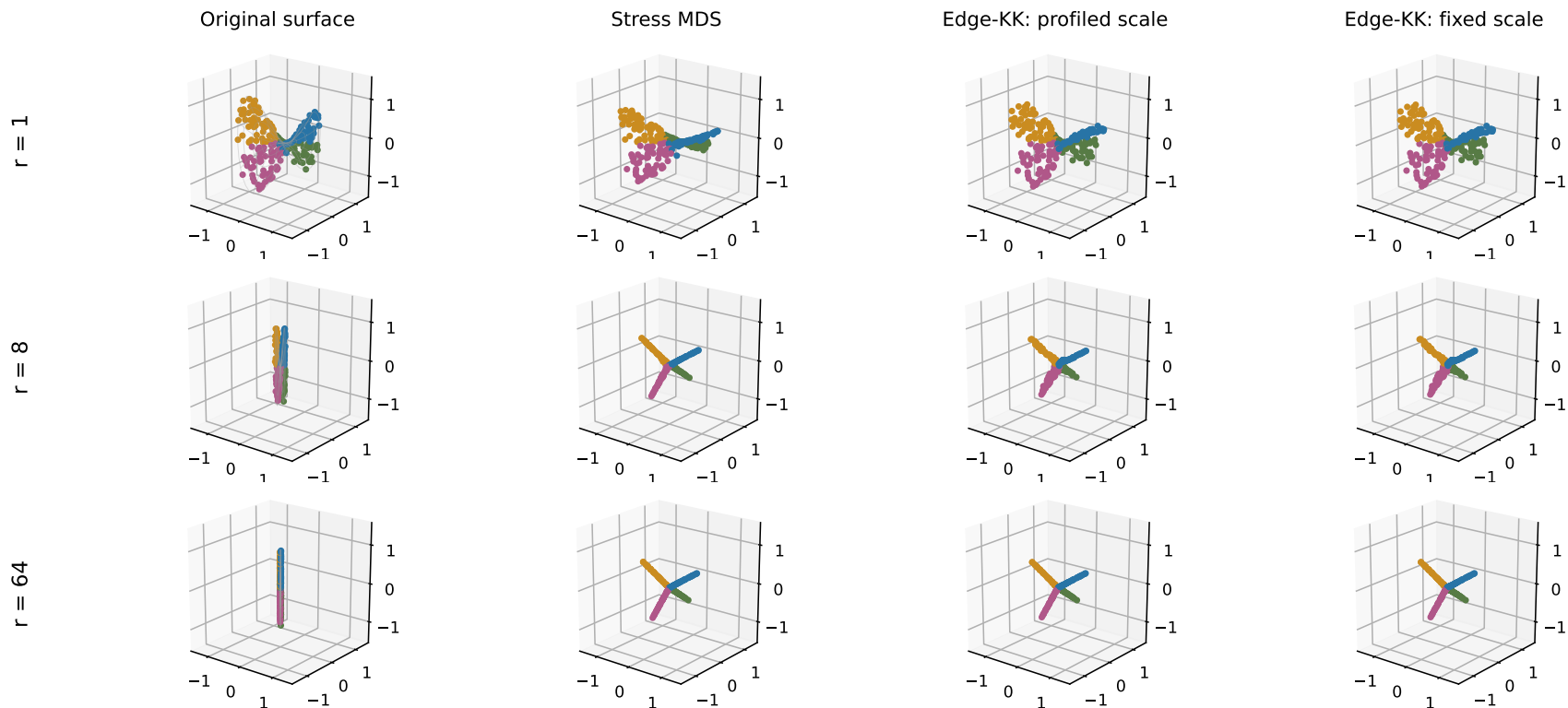


Original wireframe covers the full parameter disk. Thin equal-unit views do not mean the surface domain was truncated.

Sample 1, uniform base-disk sampling, $n=240$. Coordinates are aligned by an orthogonal transformation and translation, edge-calibrated, and divided uniformly by r^2 . All axes use equal units and common limits. The original wireframe spans the full disk. A thin equal-unit view does not truncate the original parameter domain. Colors preserve vertex correspondence.

Saddle: geodesic graph, $k = 32$

Saddle · geodesic graph · $k=32$ · first base-disk sample
Edge-calibrated physical coordinates / r^2 ; equal spatial units and common limits

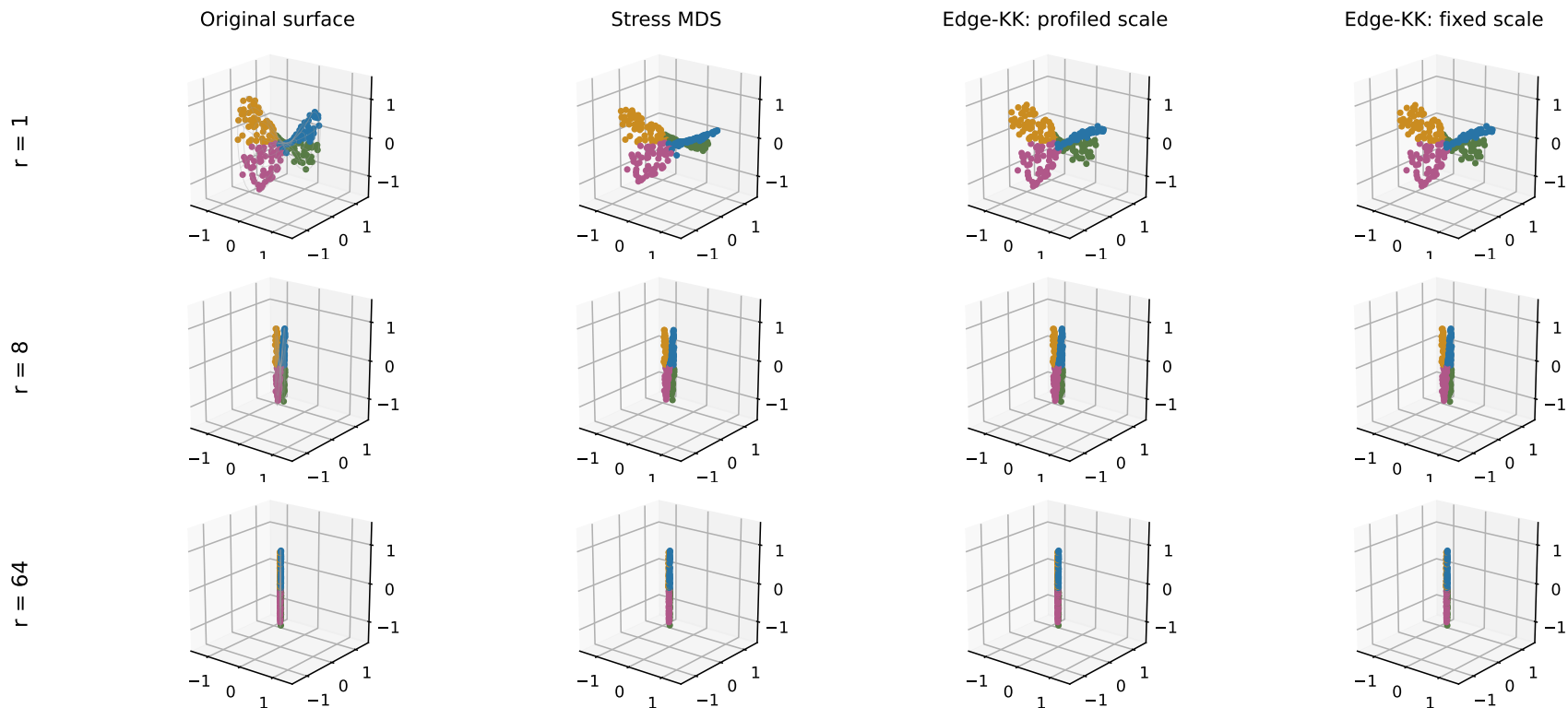


Original wireframe covers the full parameter disk. Thin equal-unit views do not mean the surface domain was truncated.

Sample 1, uniform base-disk sampling, $n=240$. Coordinates are aligned by an orthogonal transformation and translation, edge-calibrated, and divided uniformly by r^2 . All axes use equal units and common limits. The original wireframe spans the full disk. A thin equal-unit view does not truncate the original parameter domain. Colors preserve vertex correspondence.

Saddle: ambient graph, $k = 32$

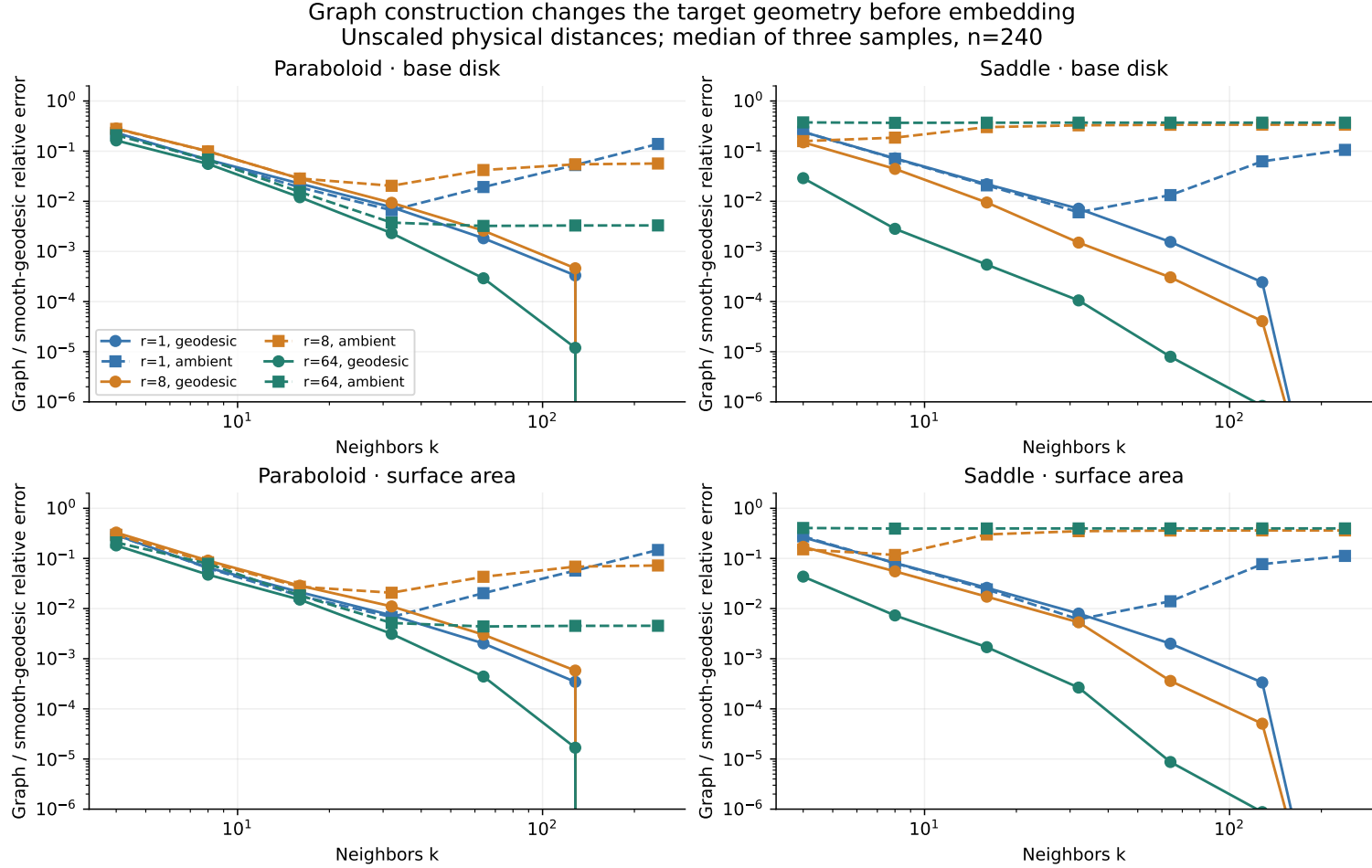
Saddle · ambient graph · $k=32$ · first base-disk sample
Edge-calibrated physical coordinates / r^2 ; equal spatial units and common limits



Original wireframe covers the full parameter disk. Thin equal-unit views do not mean the surface domain was truncated.

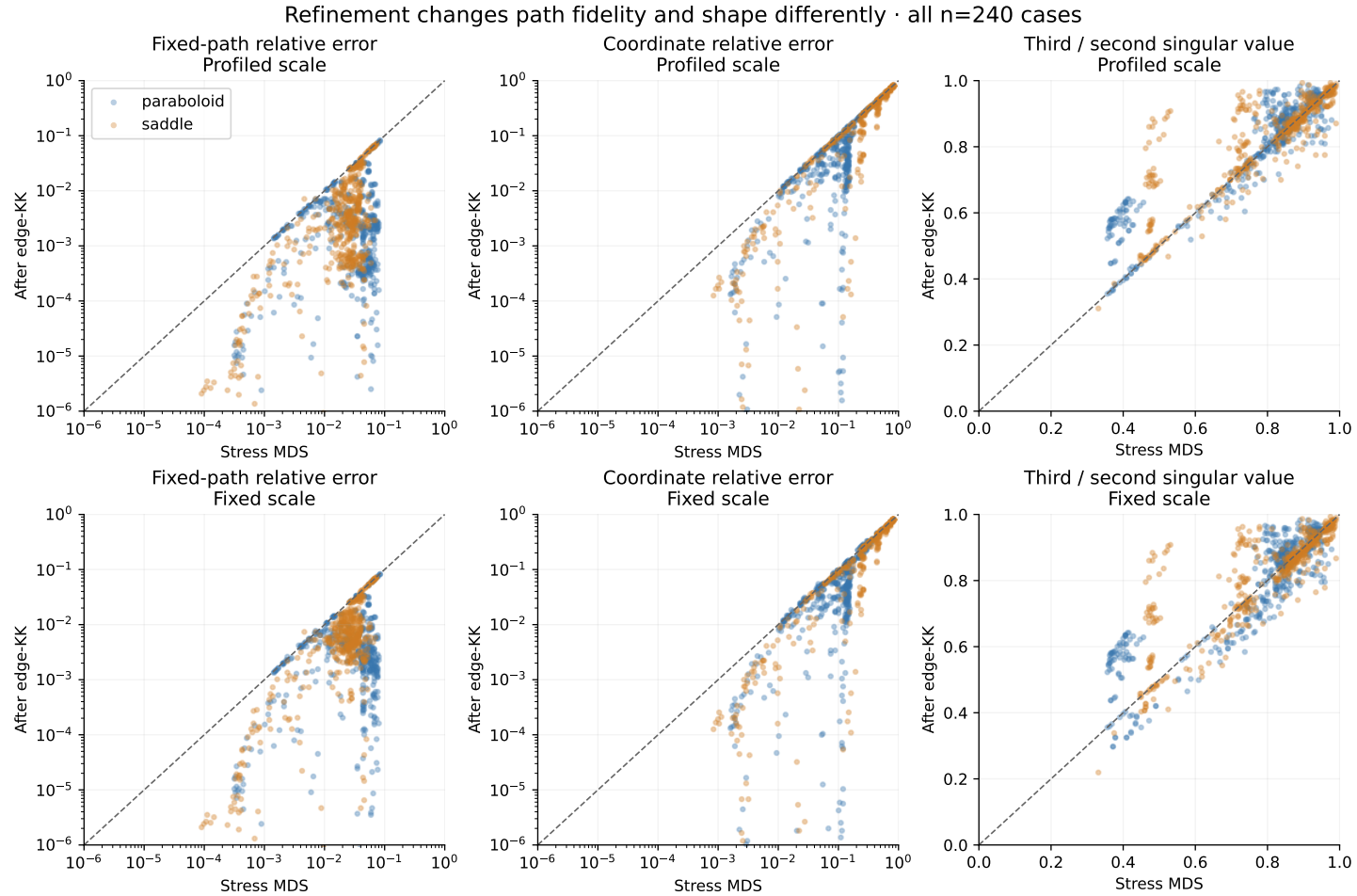
Sample 1, uniform base-disk sampling, $n=240$. Coordinates are aligned by an orthogonal transformation and translation, edge-calibrated, and divided uniformly by r^2 . All axes use equal units and common limits. The original wireframe spans the full disk. A thin equal-unit view does not truncate the original parameter domain. Colors preserve vertex correspondence.

Graph fidelity precedes embedding



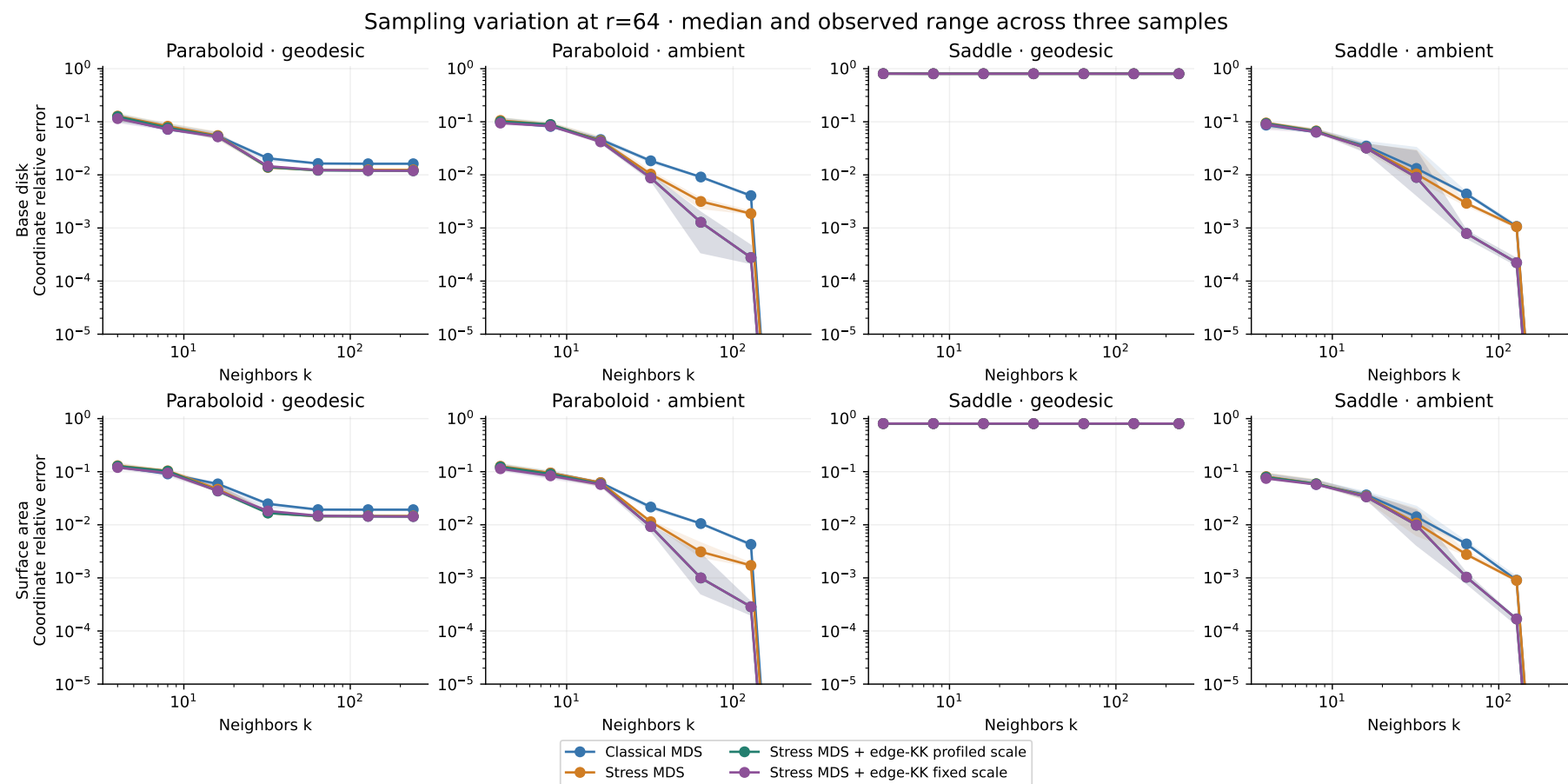
The original-coordinate row supplies one graph/reference value per graph. Physical graph distances are compared with smooth geodesics without fitting a scale. Curves show medians over three n=240 samples at representative radii. Values below 10^{-6} , including complete-geodesic numerical controls, fall outside the displayed range.

Path fidelity and geometry are different comparisons



Each point is one sample/radius/graph setting. Points below the diagonal have smaller values after refinement. Logarithmic axes clip values below 10^{-6} , including exact ambient controls; complete numerical values are in scores.csv. A lower singular-value ratio describes shape, and is not inherently a better fit.

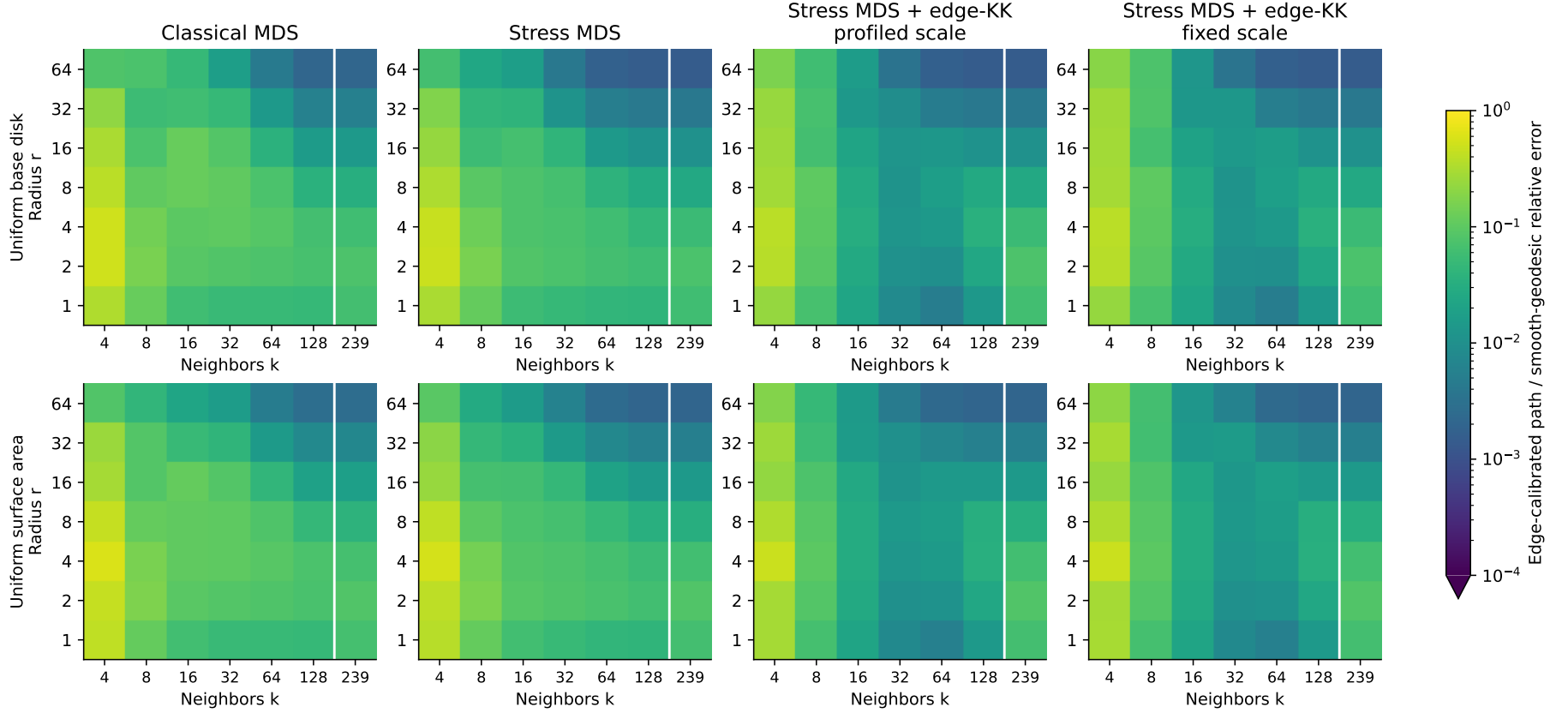
Variation across independent samples



At $r=64$, lines show the median and shaded bands the observed range over three independent samples. These ranges are not confidence intervals. Both sampling measures and graph regimes are shown. Values below 10^{-5} are outside the displayed logarithmic range, including exact complete-ambient controls.

Paraboloid, geodesic graph: path agreement with smooth geodesics

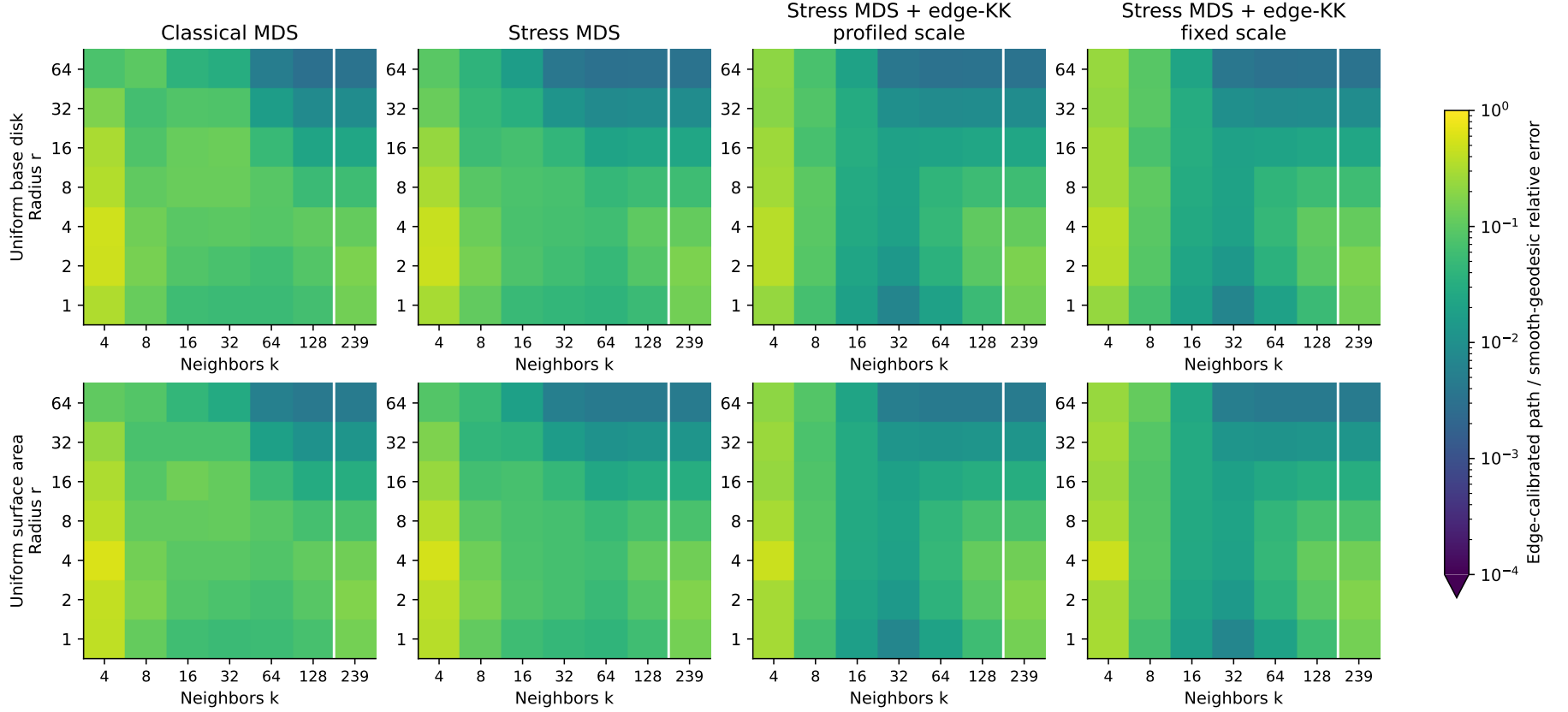
Paraboloid · geodesic neighbors and edge targets
Edge-calibrated path / smooth-geodesic relative error; median of three samples, n=240



Every cell is the median over three independent n=240 samples at the indicated radius and neighborhood size. Rows compare base-disk and surface-area sampling; columns compare initializers and scale controls. The separated final column is the complete graph, k=239. Color scales are shared across methods and measures. Values below 10^{-4} use the lowest color; the exact values are retained in the tables.

Paraboloid, ambient graph: path agreement with smooth geodesics

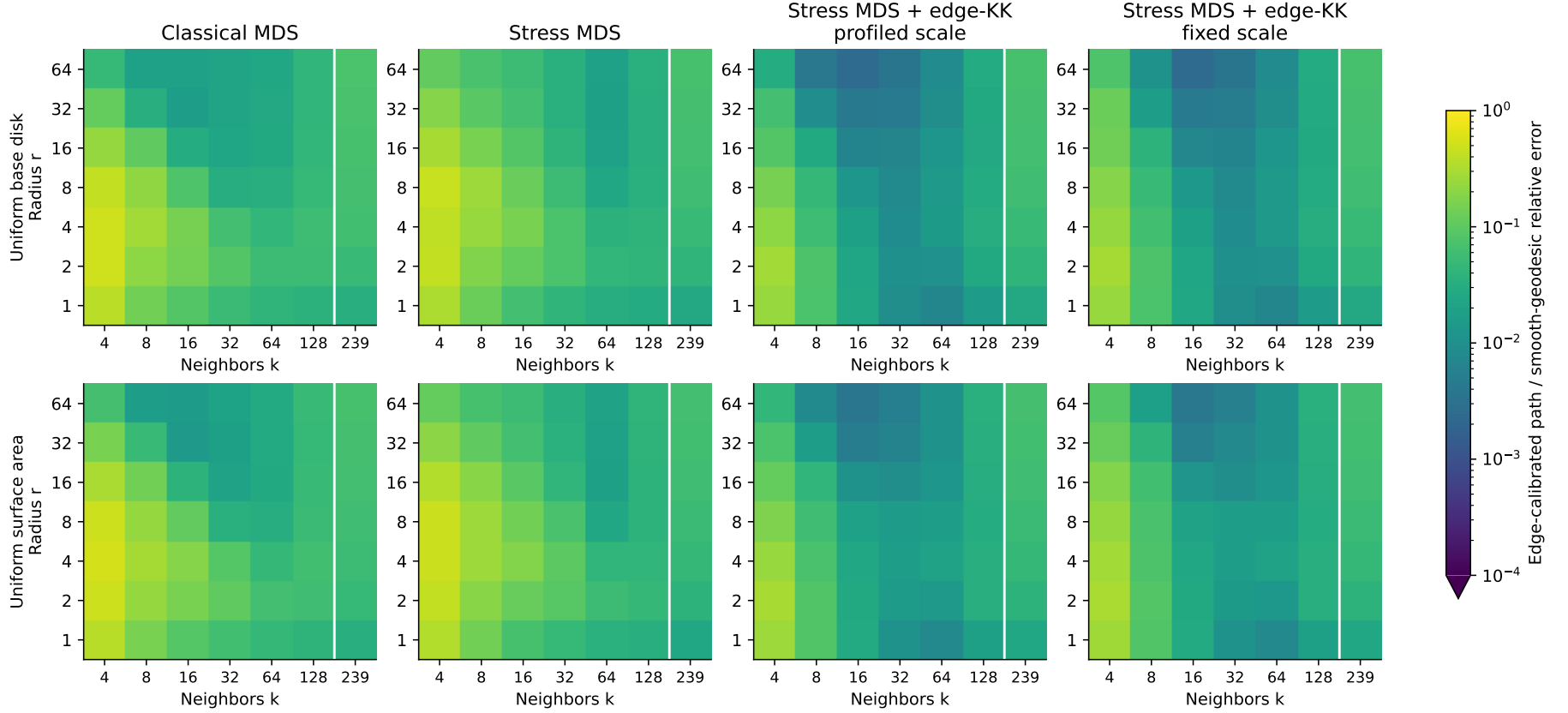
Paraboloid · ambient neighbors and edge targets
Edge-calibrated path / smooth-geodesic relative error; median of three samples, $n=240$



Every cell is the median over three independent $n=240$ samples at the indicated radius and neighborhood size. Rows compare base-disk and surface-area sampling; columns compare initializers and scale controls. The separated final column is the complete graph, $k=239$. Color scales are shared across methods and measures. Values below 10^{-4} use the lowest color; the exact values are retained in the tables.

Saddle, geodesic graph: path agreement with smooth geodesics

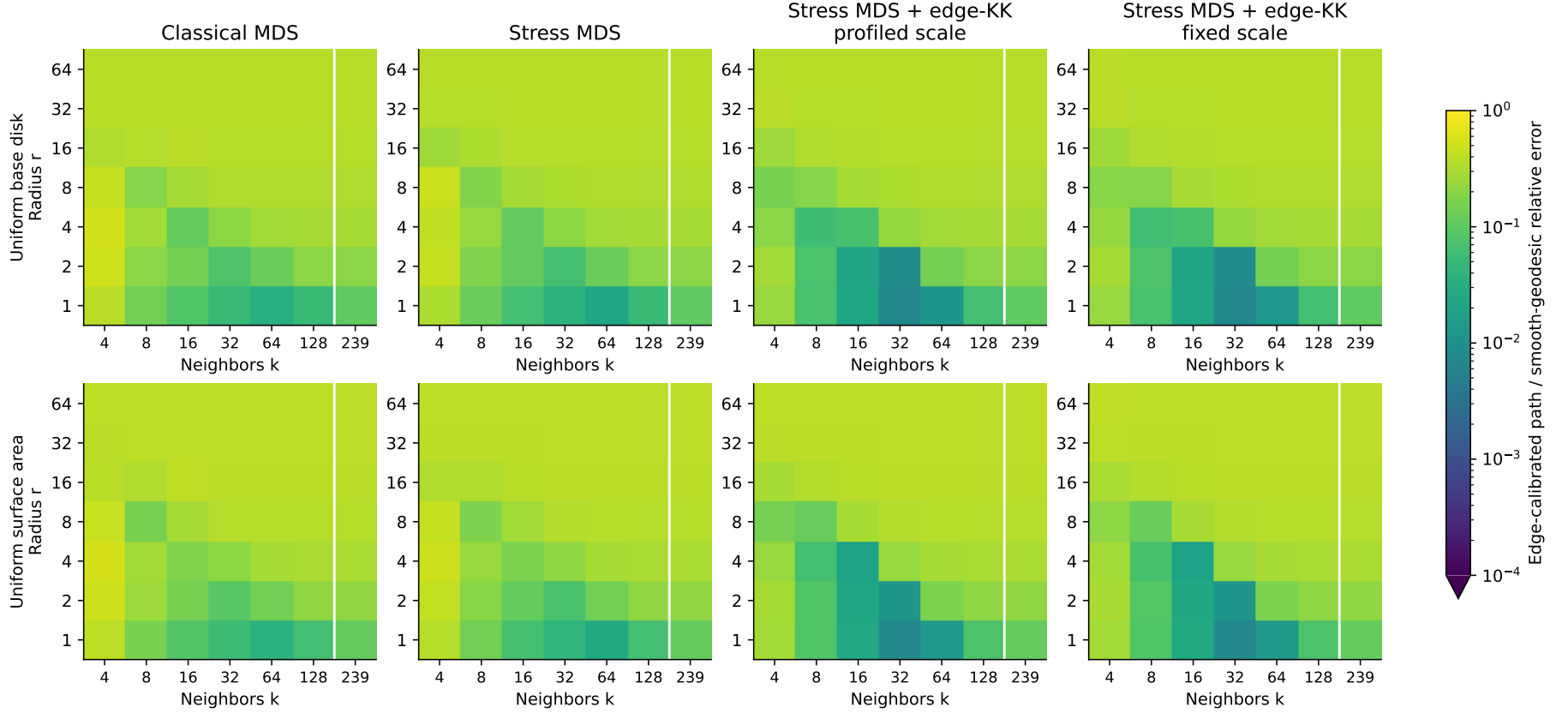
Saddle · geodesic neighbors and edge targets
Edge-calibrated path / smooth-geodesic relative error; median of three samples, $n=240$



Every cell is the median over three independent $n=240$ samples at the indicated radius and neighborhood size. Rows compare base-disk and surface-area sampling; columns compare initializers and scale controls. The separated final column is the complete graph, $k=239$. Color scales are shared across methods and measures. Values below 10^{-4} use the lowest color; the exact values are retained in the tables.

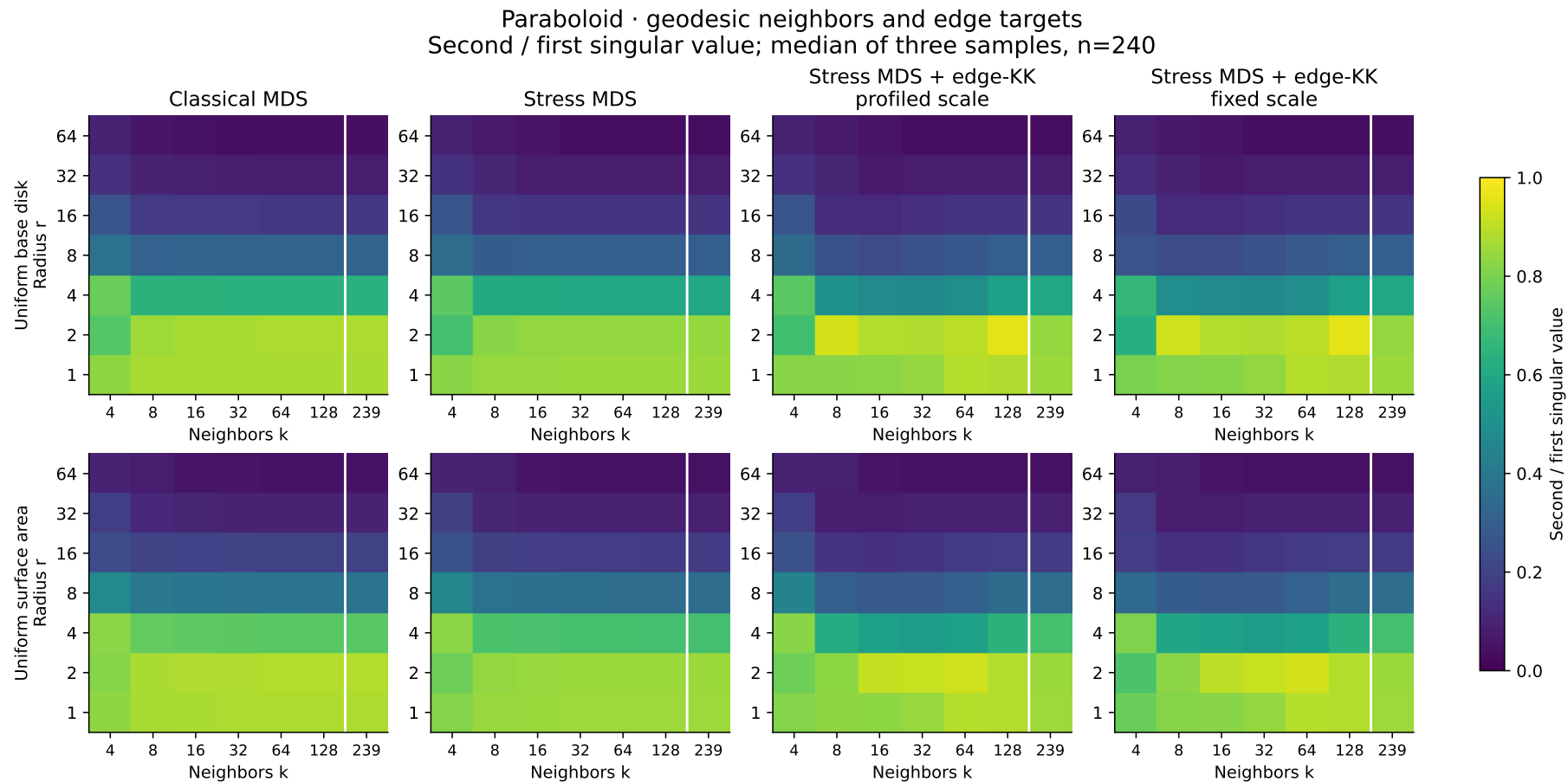
Saddle, ambient graph: path agreement with smooth geodesics

Saddle · ambient neighbors and edge targets
Edge-calibrated path / smooth-geodesic relative error; median of three samples, $n=240$



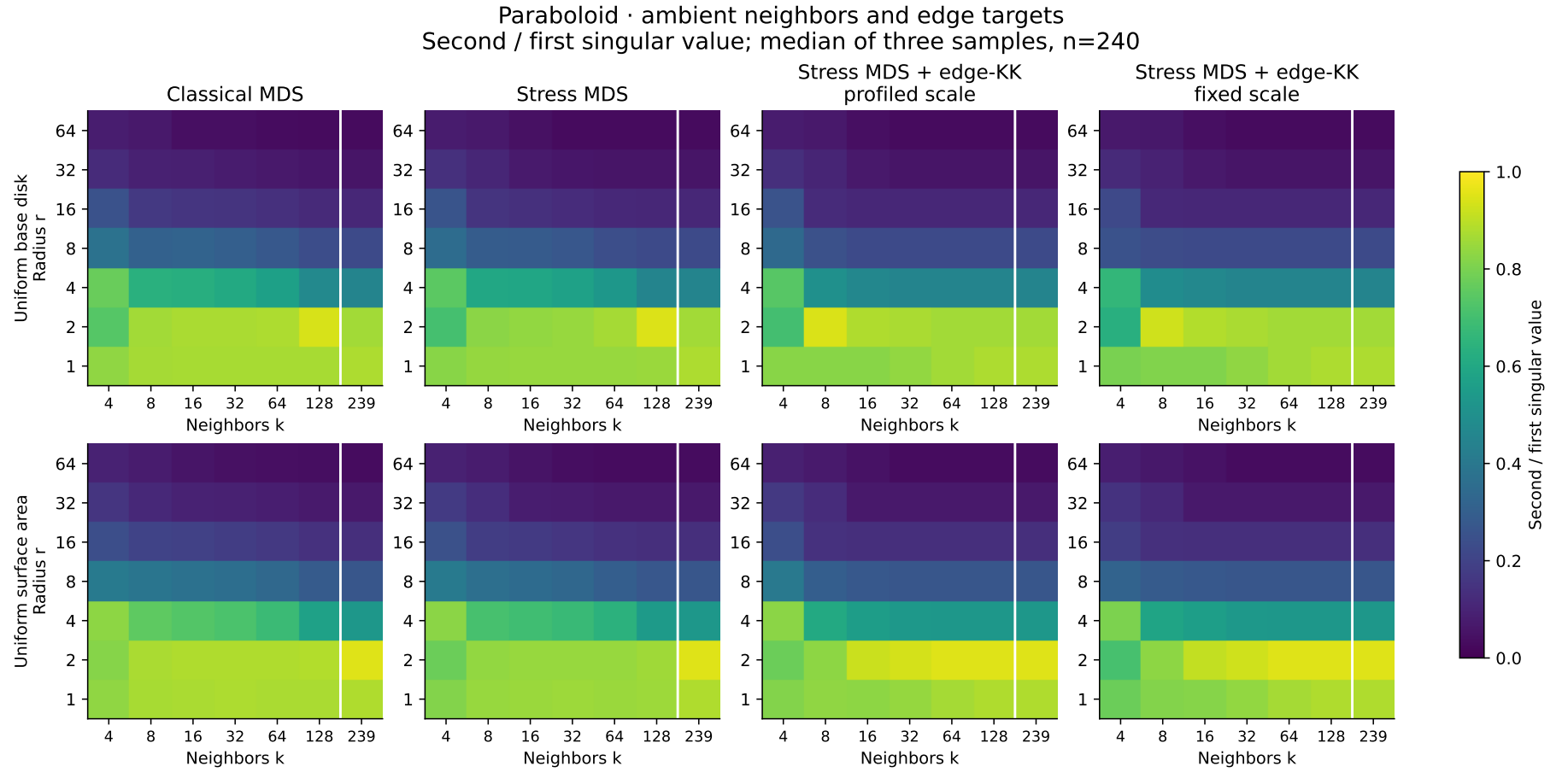
Every cell is the median over three independent $n=240$ samples at the indicated radius and neighborhood size. Rows compare base-disk and surface-area sampling; columns compare initializers and scale controls. The separated final column is the complete graph, $k=239$. Color scales are shared across methods and measures. Values below 10^{-4} use the lowest color; the exact values are retained in the tables.

Paraboloid, geodesic graph: elongation



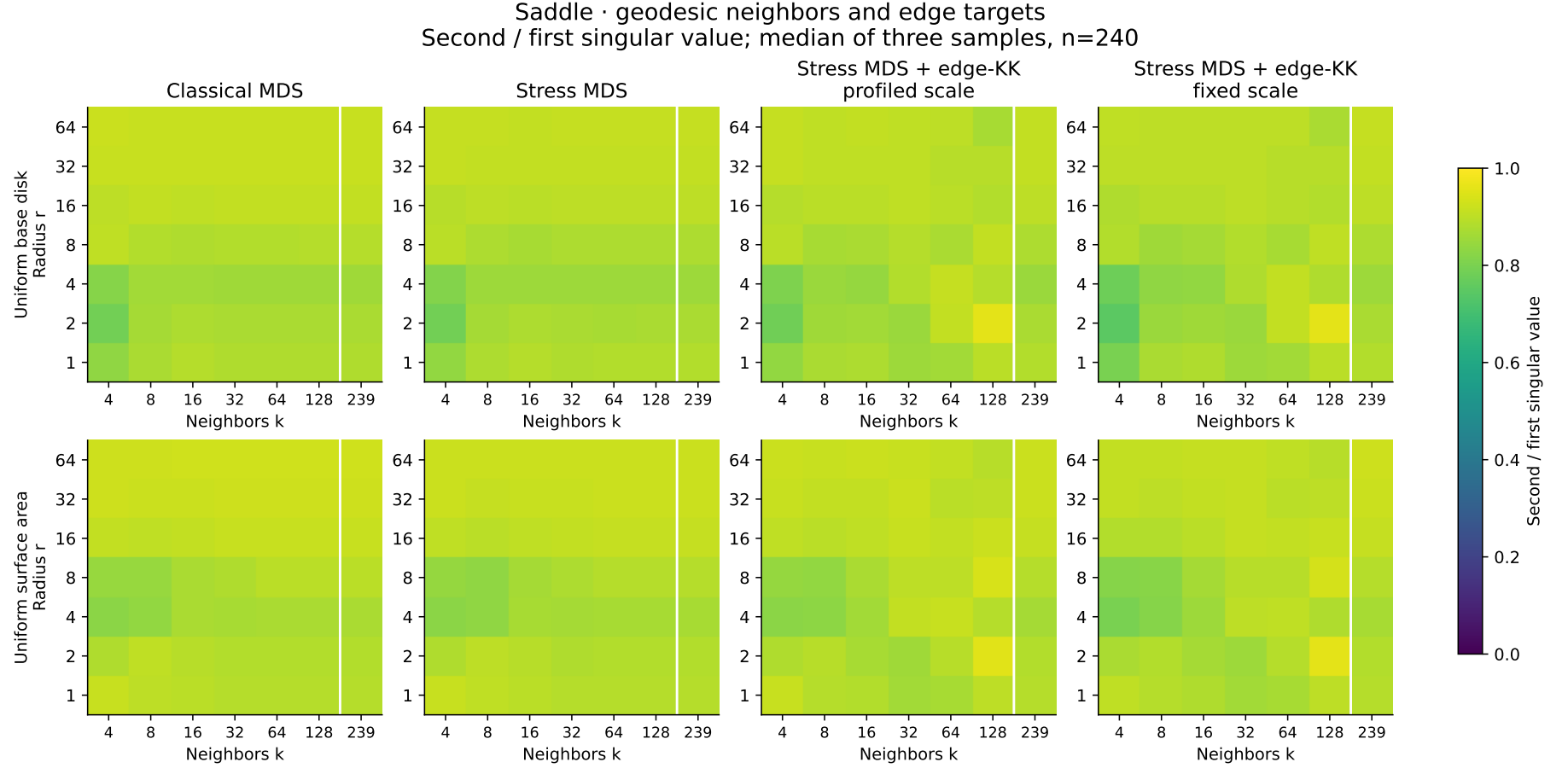
Every cell is the median over three independent n=240 samples at the indicated radius and neighborhood size. Rows compare base-disk and surface-area sampling; columns compare initializers and scale controls. The separated final column is the complete graph, k=239. Color scales are shared across methods and measures. Both singular-value ratios must be read together to distinguish a line from a plane.

Paraboloid, ambient graph: elongation



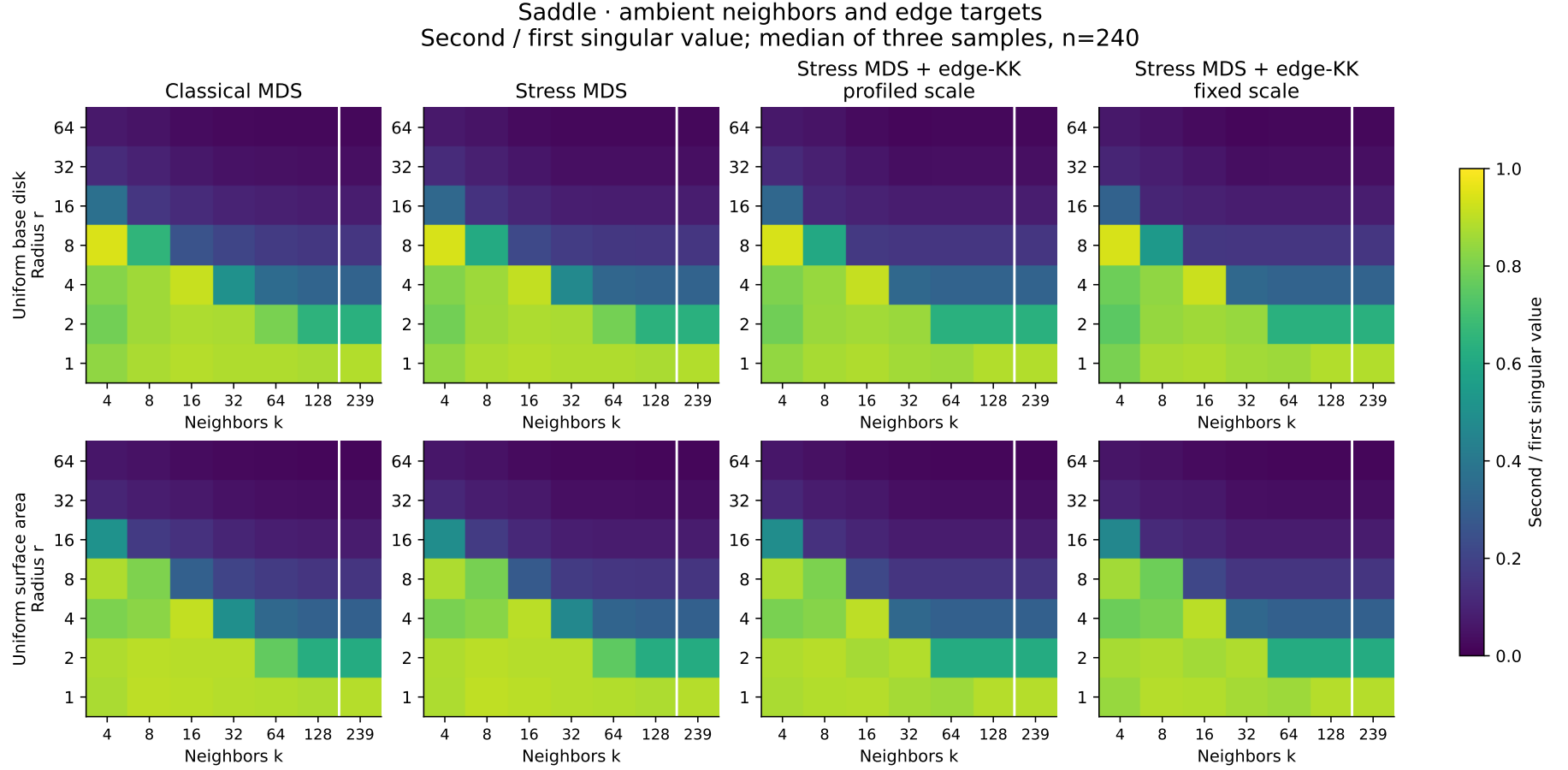
Every cell is the median over three independent n=240 samples at the indicated radius and neighborhood size. Rows compare base-disk and surface-area sampling; columns compare initializers and scale controls. The separated final column is the complete graph, k=239. Color scales are shared across methods and measures. Both singular-value ratios must be read together to distinguish a line from a plane.

Saddle, geodesic graph: elongation



Every cell is the median over three independent n=240 samples at the indicated radius and neighborhood size. Rows compare base-disk and surface-area sampling; columns compare initializers and scale controls. The separated final column is the complete graph, k=239. Color scales are shared across methods and measures. Both singular-value ratios must be read together to distinguish a line from a plane.

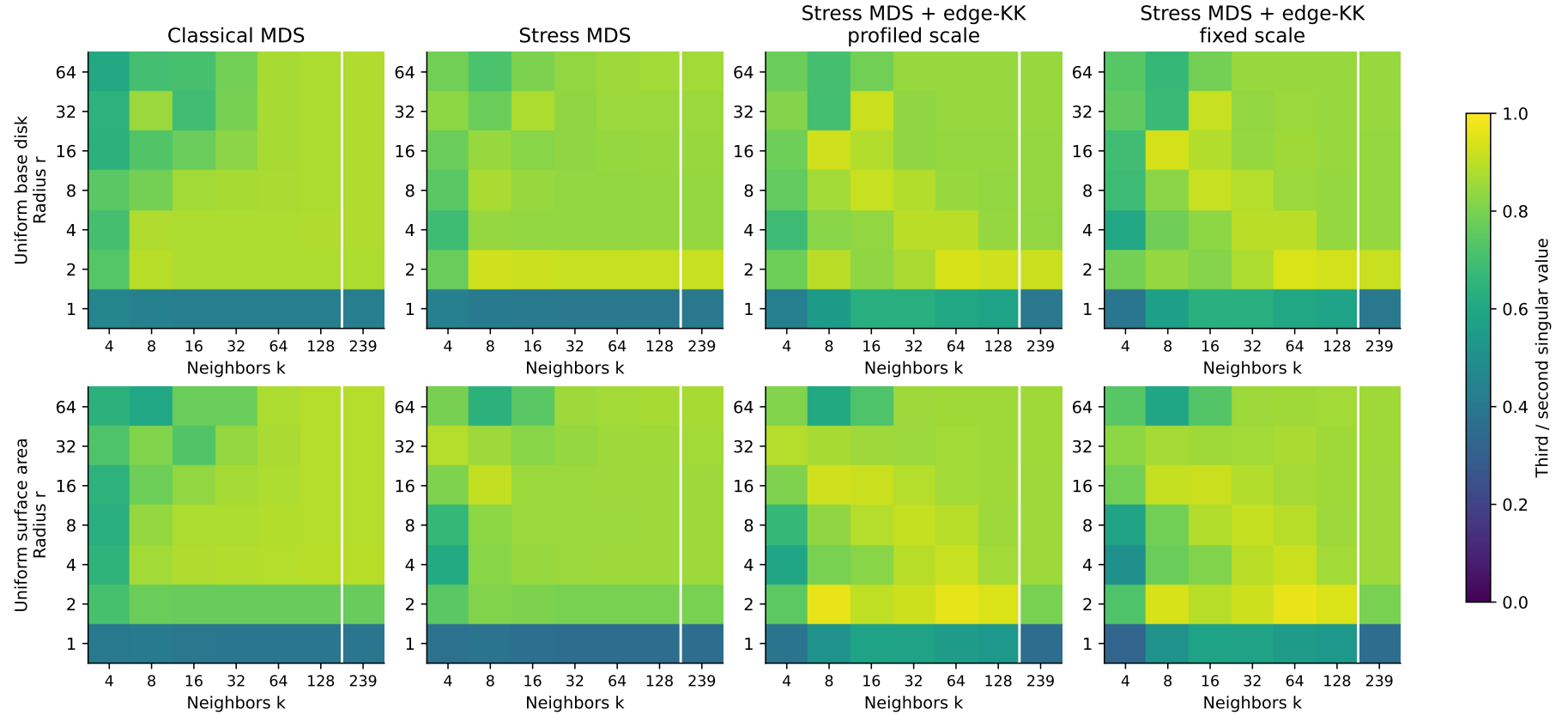
Saddle, ambient graph: elongation



Every cell is the median over three independent n=240 samples at the indicated radius and neighborhood size. Rows compare base-disk and surface-area sampling; columns compare initializers and scale controls. The separated final column is the complete graph, k=239. Color scales are shared across methods and measures. Both singular-value ratios must be read together to distinguish a line from a plane.

Paraboloid, geodesic graph: planarity

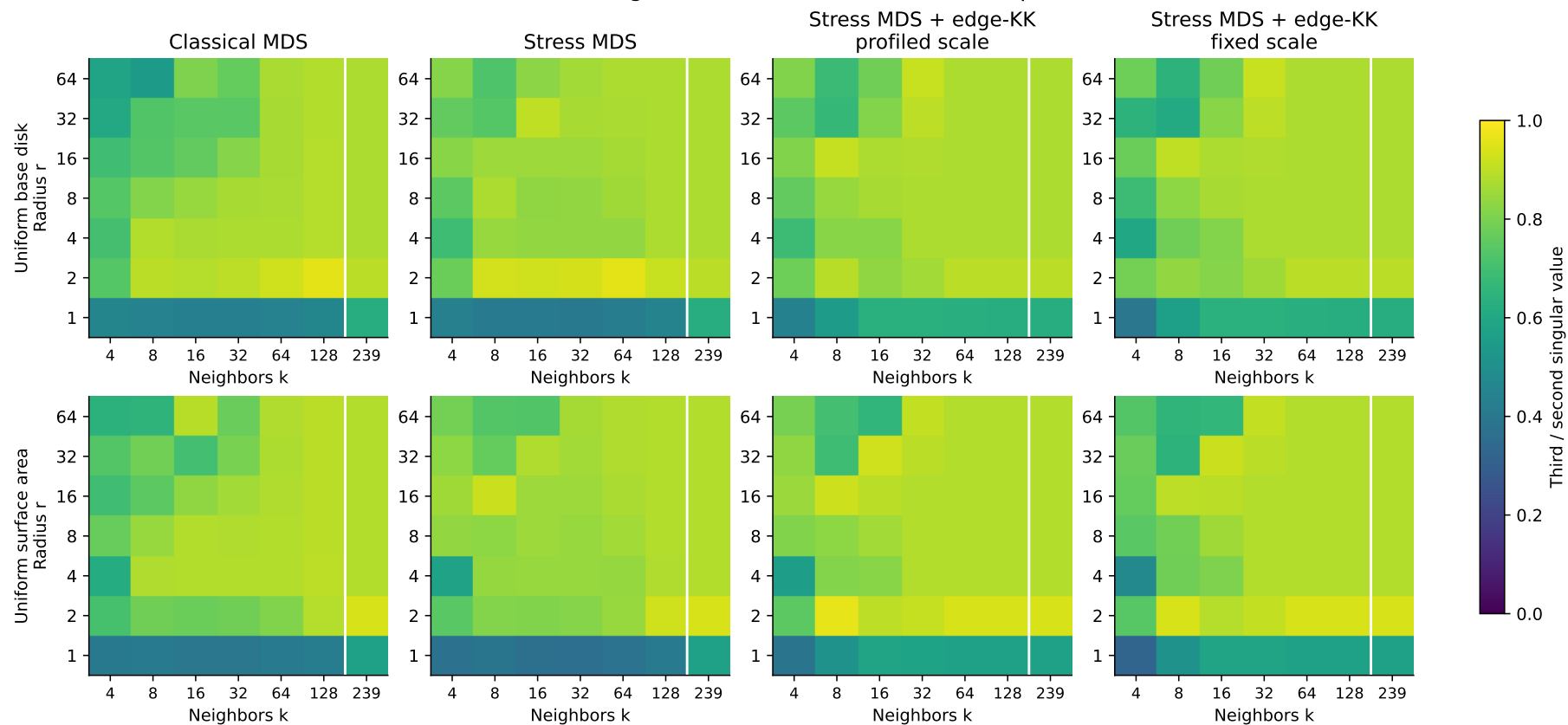
Paraboloid · geodesic neighbors and edge targets
Third / second singular value; median of three samples, n=240



Every cell is the median over three independent n=240 samples at the indicated radius and neighborhood size. Rows compare base-disk and surface-area sampling; columns compare initializers and scale controls. The separated final column is the complete graph, k=239. Color scales are shared across methods and measures. Both singular-value ratios must be read together to distinguish a line from a plane.

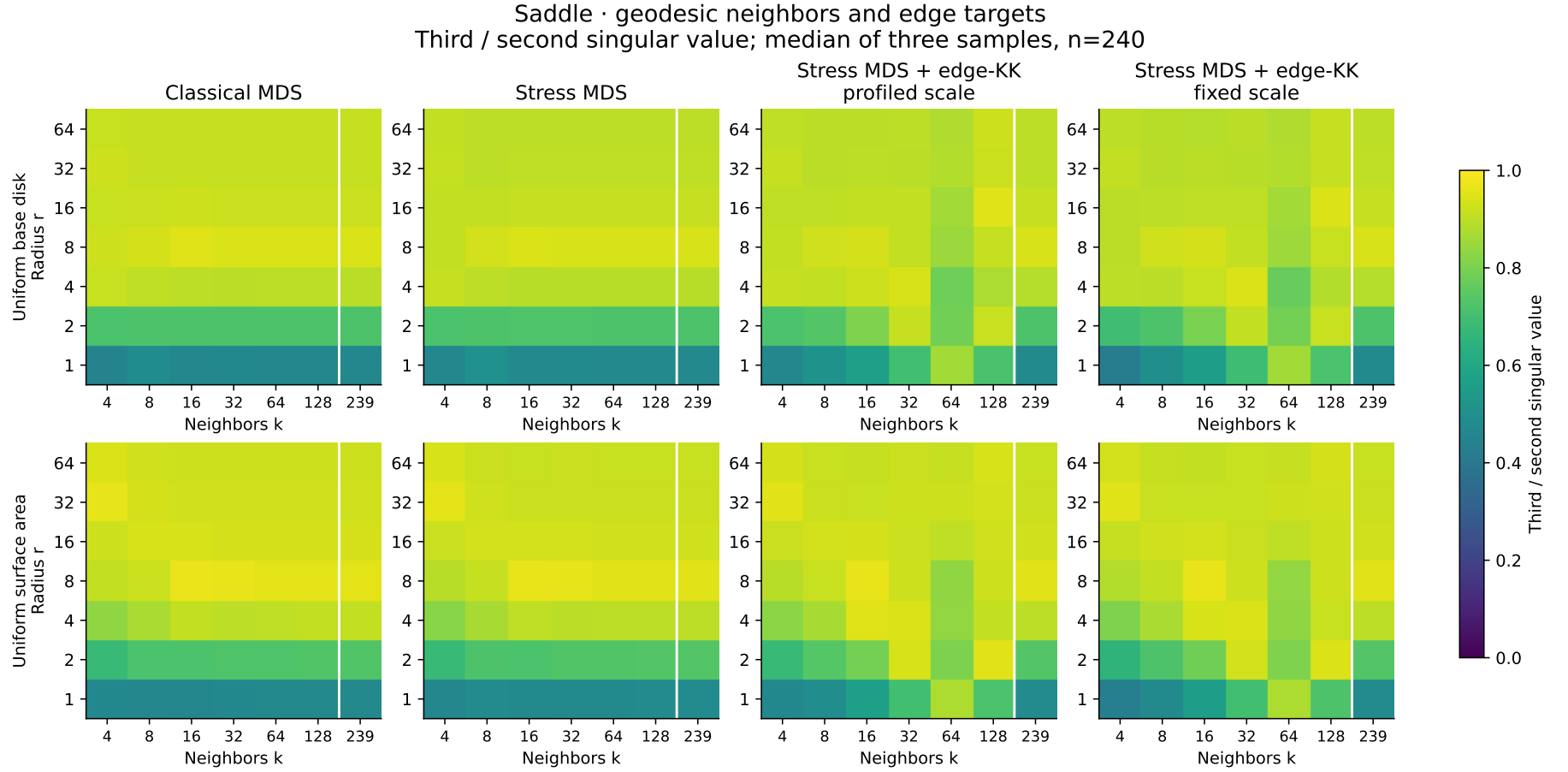
Paraboloid, ambient graph: planarity

Paraboloid · ambient neighbors and edge targets
Third / second singular value; median of three samples, n=240



Every cell is the median over three independent n=240 samples at the indicated radius and neighborhood size. Rows compare base-disk and surface-area sampling; columns compare initializers and scale controls. The separated final column is the complete graph, k=239. Color scales are shared across methods and measures. Both singular-value ratios must be read together to distinguish a line from a plane.

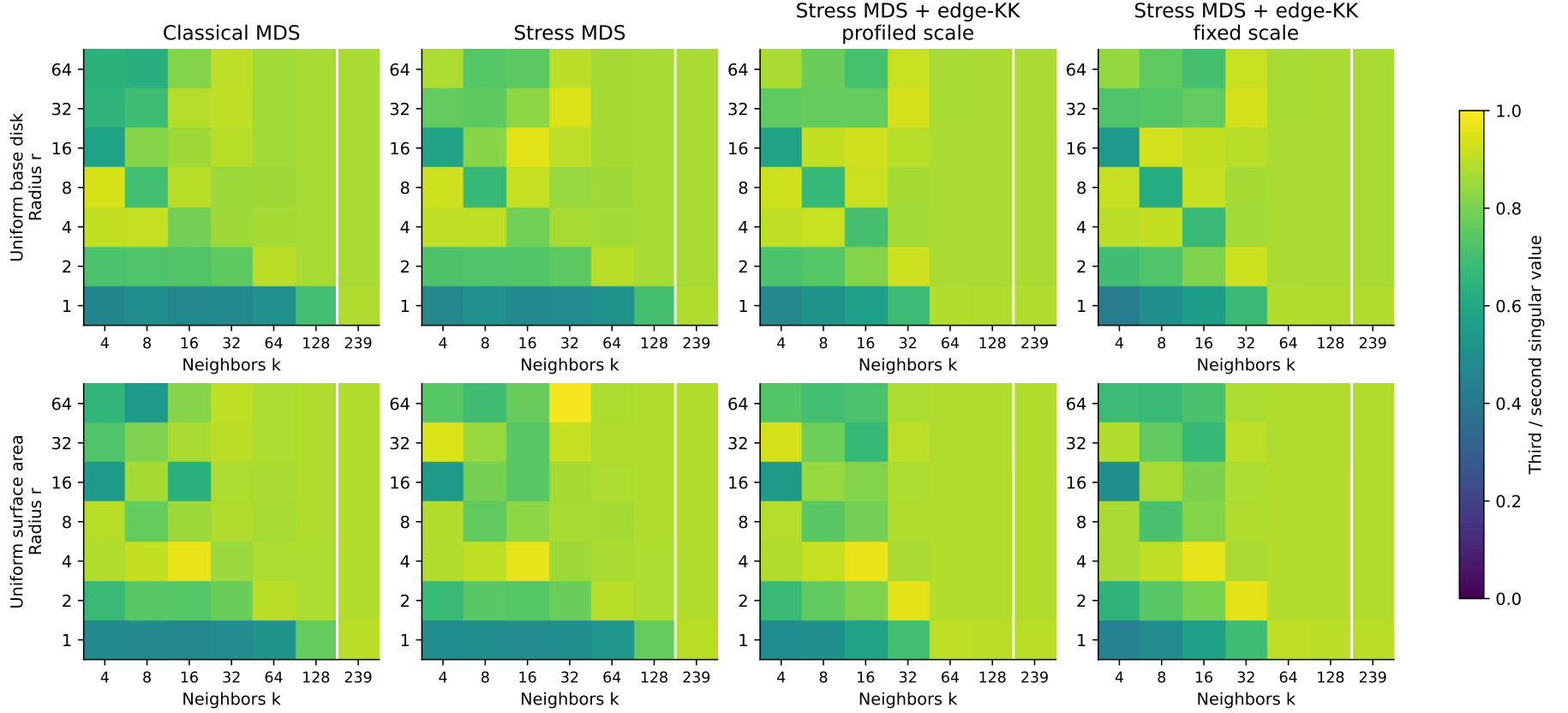
Saddle, geodesic graph: planarity



Every cell is the median over three independent n=240 samples at the indicated radius and neighborhood size. Rows compare base-disk and surface-area sampling; columns compare initializers and scale controls. The separated final column is the complete graph, k=239. Color scales are shared across methods and measures. Both singular-value ratios must be read together to distinguish a line from a plane.

Saddle, ambient graph: planarity

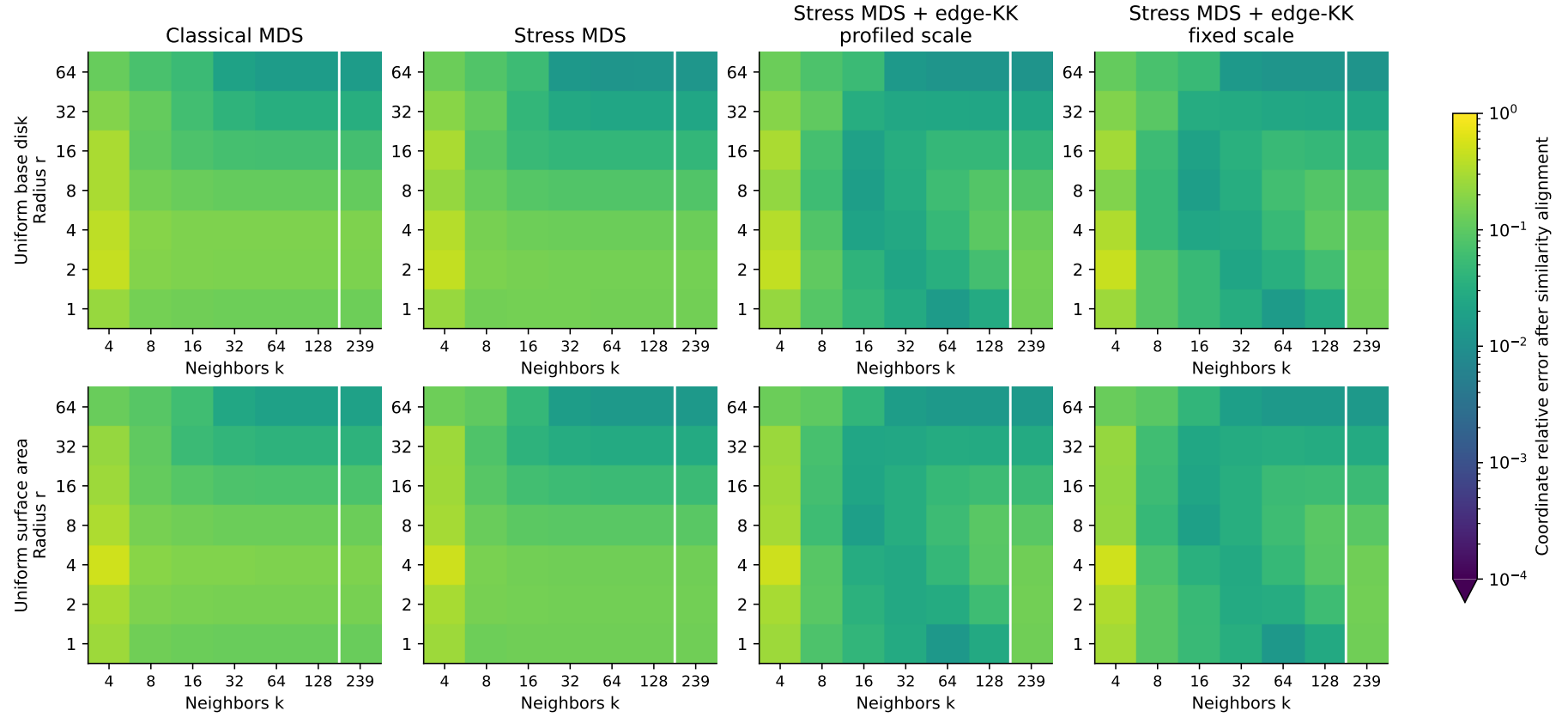
Saddle · ambient neighbors and edge targets
Third / second singular value; median of three samples, n=240



Every cell is the median over three independent n=240 samples at the indicated radius and neighborhood size. Rows compare base-disk and surface-area sampling; columns compare initializers and scale controls. The separated final column is the complete graph, k=239. Color scales are shared across methods and measures. Both singular-value ratios must be read together to distinguish a line from a plane.

Paraboloid, geodesic graph: coordinate agreement

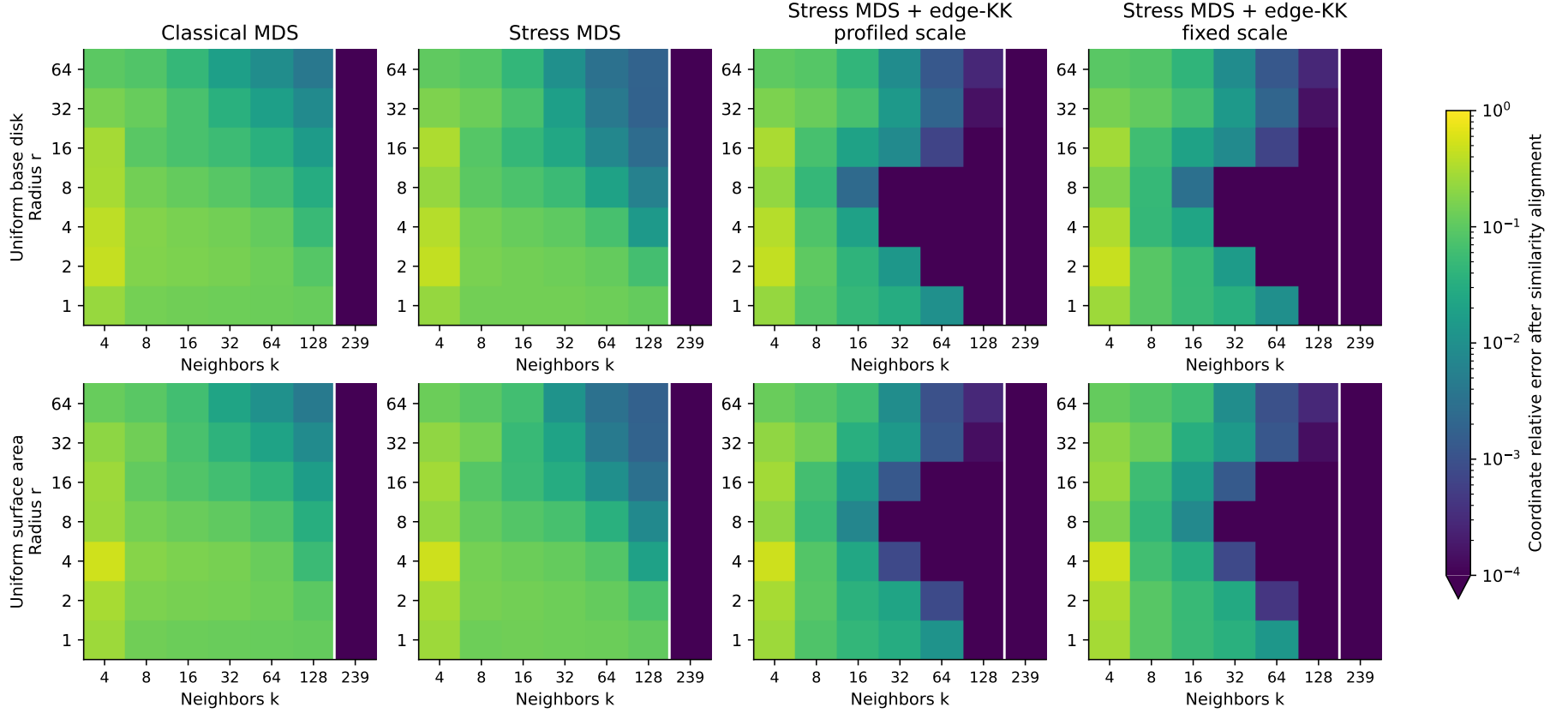
Paraboloid · geodesic neighbors and edge targets
Coordinate relative error after similarity alignment; median of three samples, n=240



Every cell is the median over three independent n=240 samples at the indicated radius and neighborhood size. Rows compare base-disk and surface-area sampling; columns compare initializers and scale controls. The separated final column is the complete graph, k=239. Color scales are shared across methods and measures. Values below 10^{-4} use the lowest color; the exact values are retained in the tables.

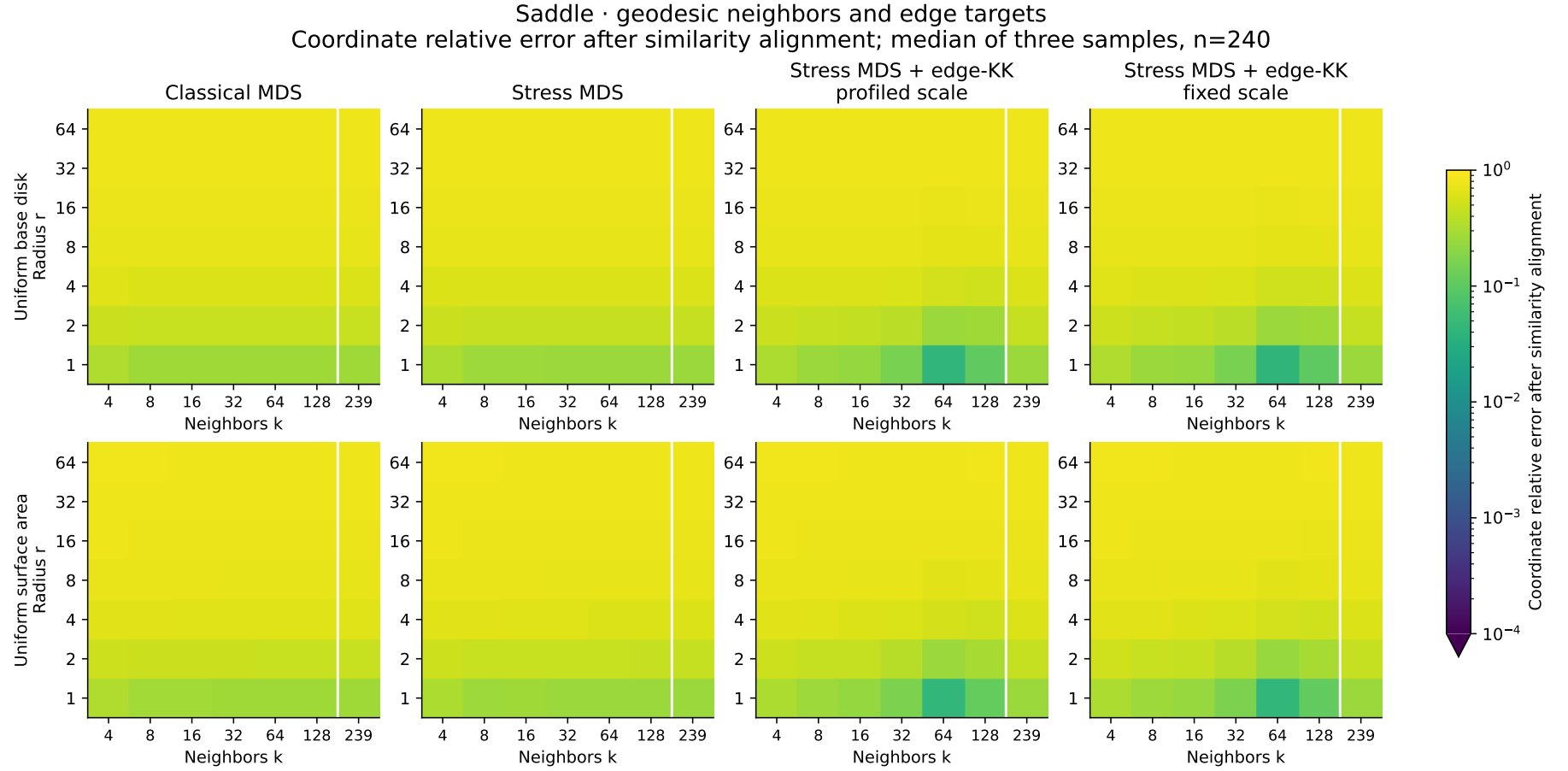
Paraboloid, ambient graph: coordinate agreement

Paraboloid · ambient neighbors and edge targets
Coordinate relative error after similarity alignment; median of three samples, n=240



Every cell is the median over three independent n=240 samples at the indicated radius and neighborhood size. Rows compare base-disk and surface-area sampling; columns compare initializers and scale controls. The separated final column is the complete graph, k=239. Color scales are shared across methods and measures. Values below 10^{-4} use the lowest color; the exact values are retained in the tables.

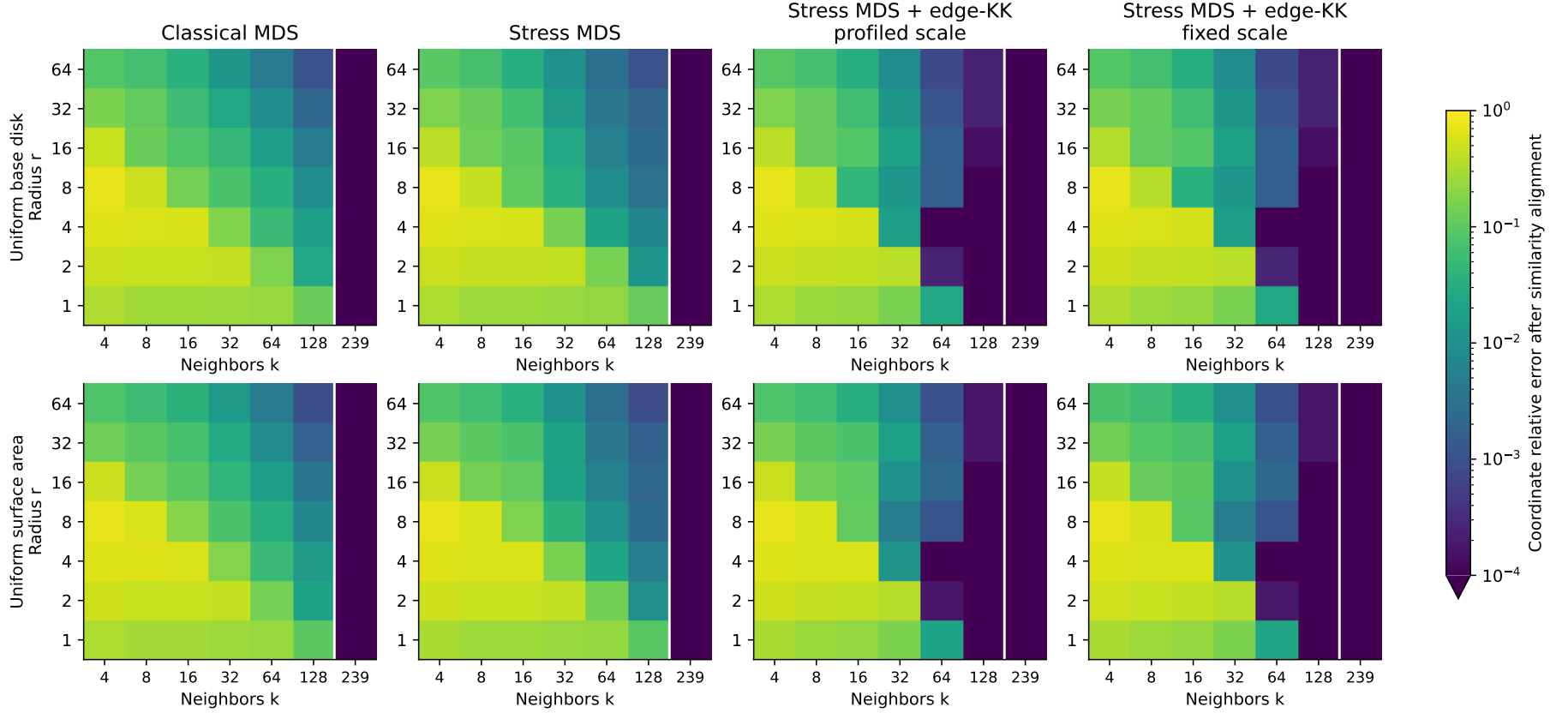
Saddle, geodesic graph: coordinate agreement



Every cell is the median over three independent n=240 samples at the indicated radius and neighborhood size. Rows compare base-disk and surface-area sampling; columns compare initializers and scale controls. The separated final column is the complete graph, k=239. Color scales are shared across methods and measures. Values below 10^{-4} use the lowest color; the exact values are retained in the tables.

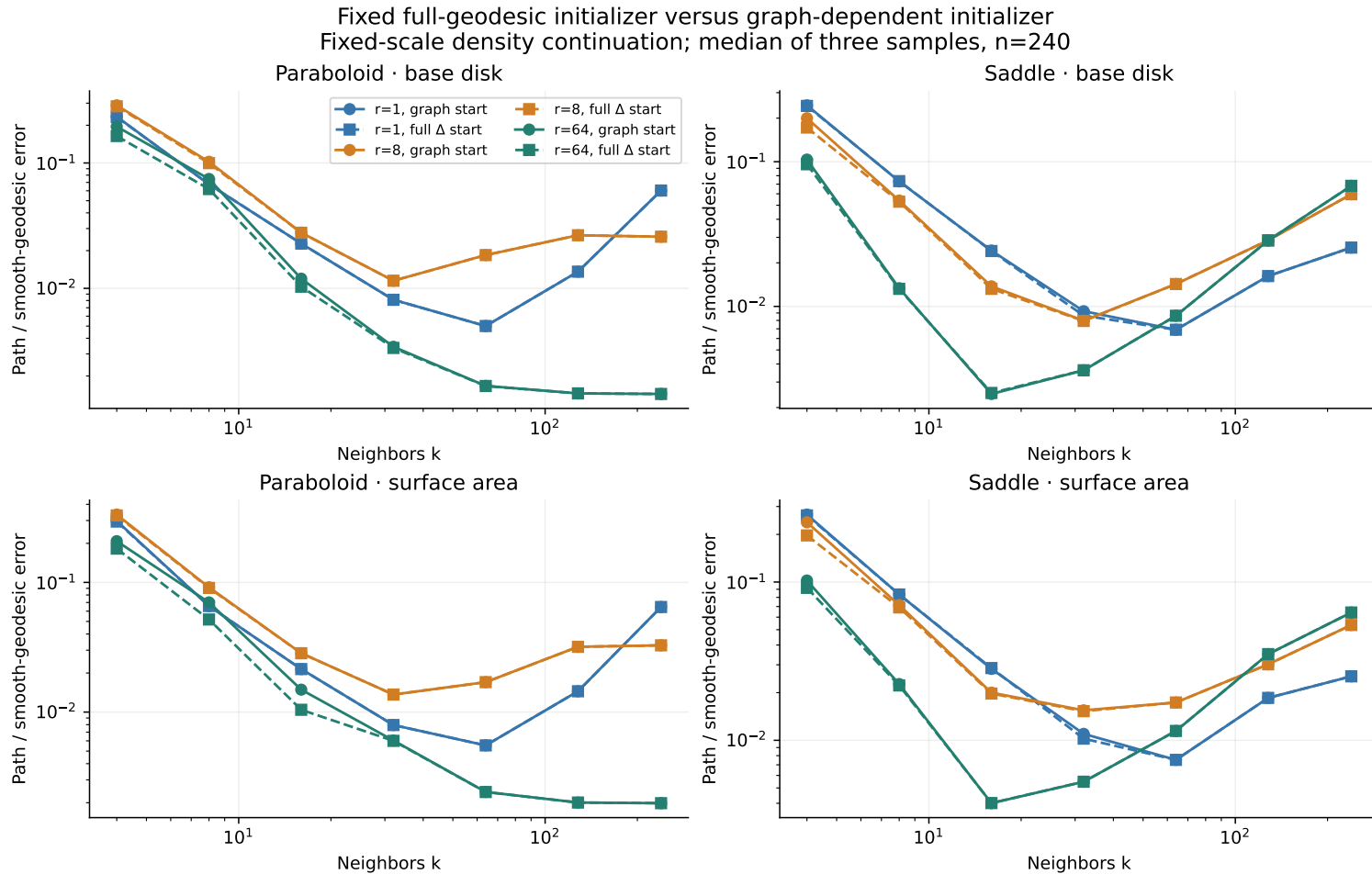
Saddle, ambient graph: coordinate agreement

Saddle · ambient neighbors and edge targets
Coordinate relative error after similarity alignment; median of three samples, $n=240$



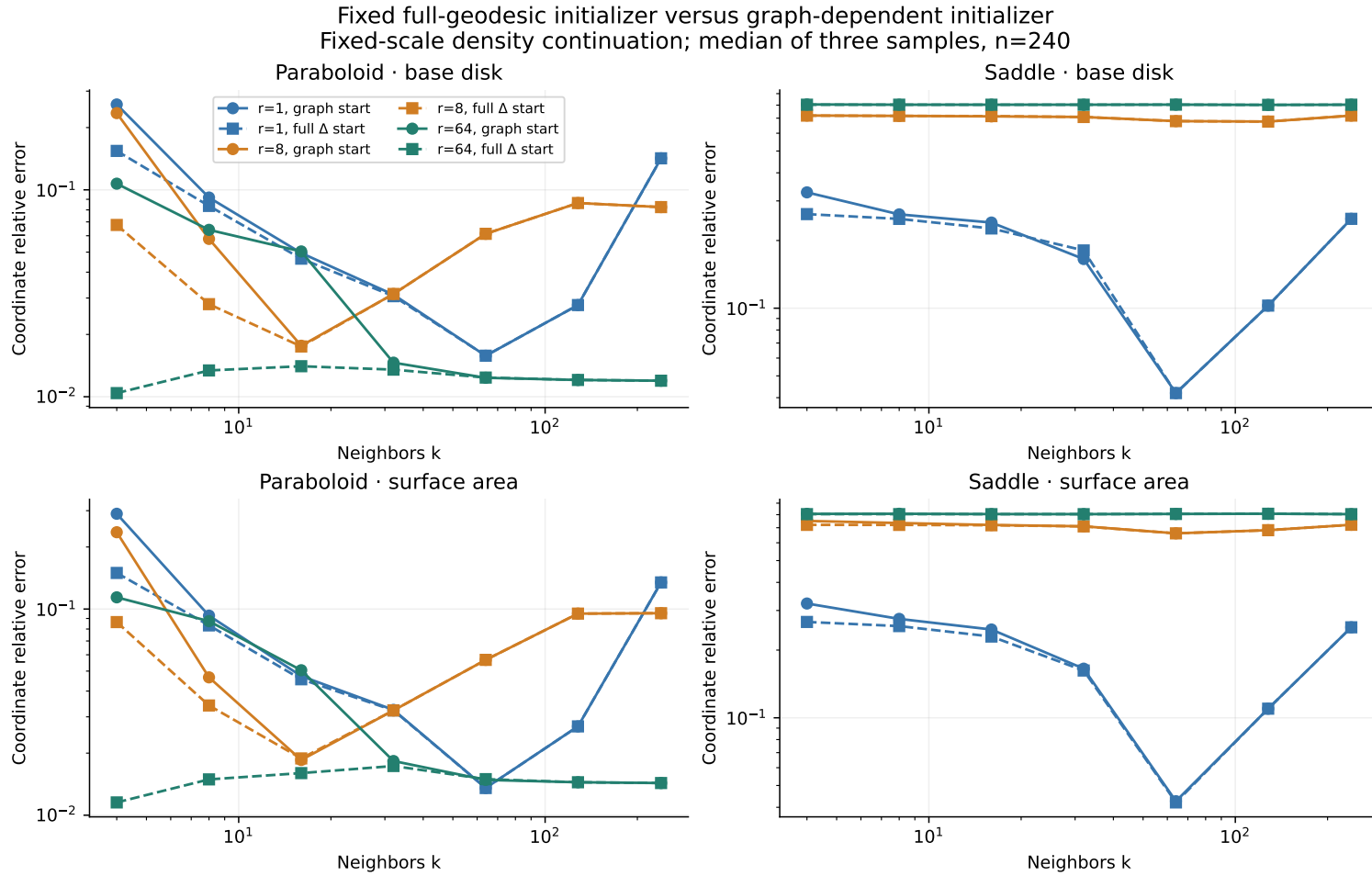
Every cell is the median over three independent $n=240$ samples at the indicated radius and neighborhood size. Rows compare base-disk and surface-area sampling; columns compare initializers and scale controls. The separated final column is the complete graph, $k=239$. Color scales are shared across methods and measures. Values below 10^{-4} use the lowest color; the exact values are retained in the tables.

Isolating the graph from the initializer



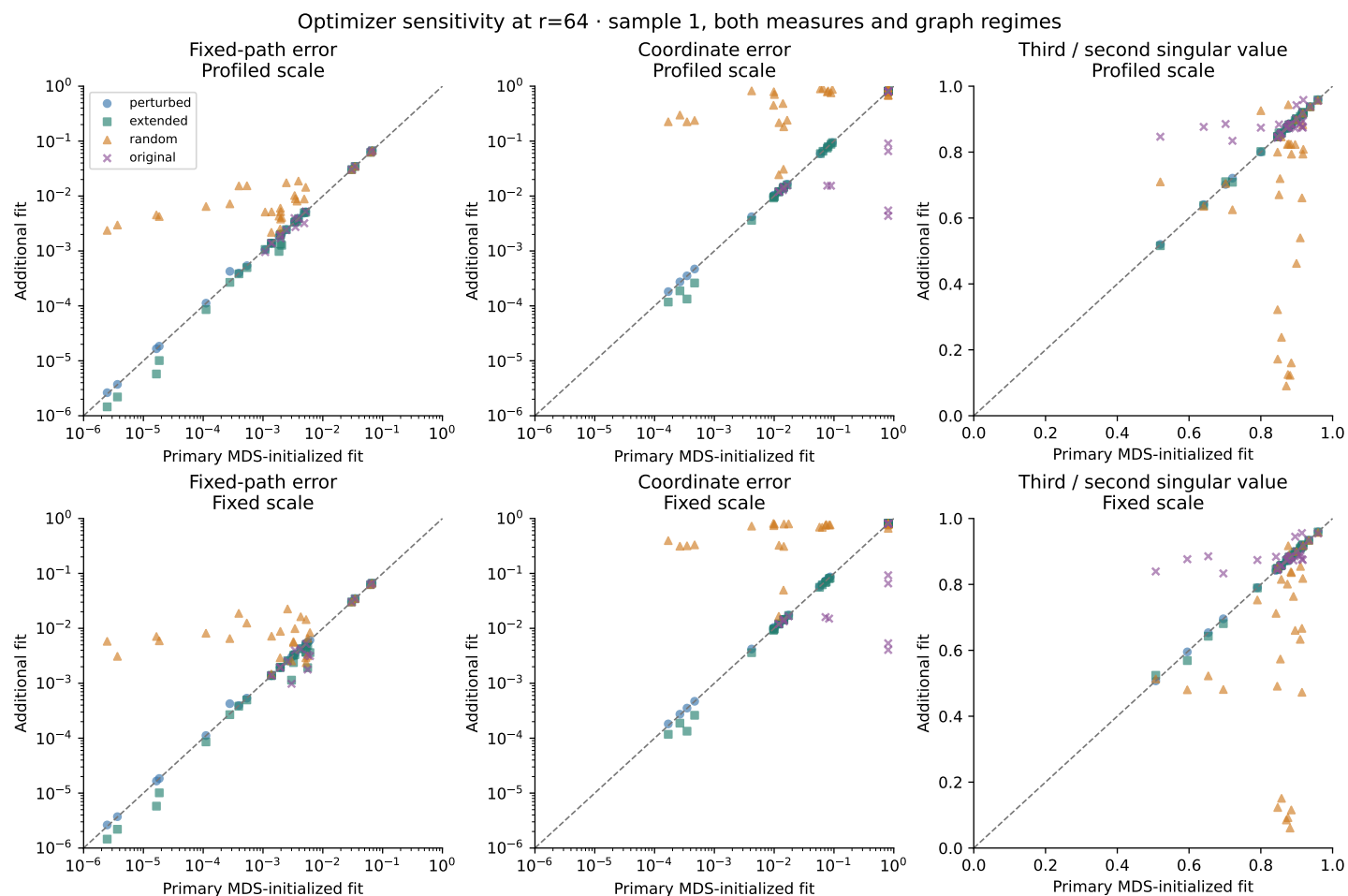
Solid curves initialize from graph distances; dashed curves reuse full-geodesic classical MDS across k at each radius and sample. Both use fixed-scale density continuation. Medians over three samples; both measures are shown. Panels use separate logarithmic y ranges. Changing k can still alter the refined geometry even when the starting configuration is fixed.

Isolating the graph from the initializer



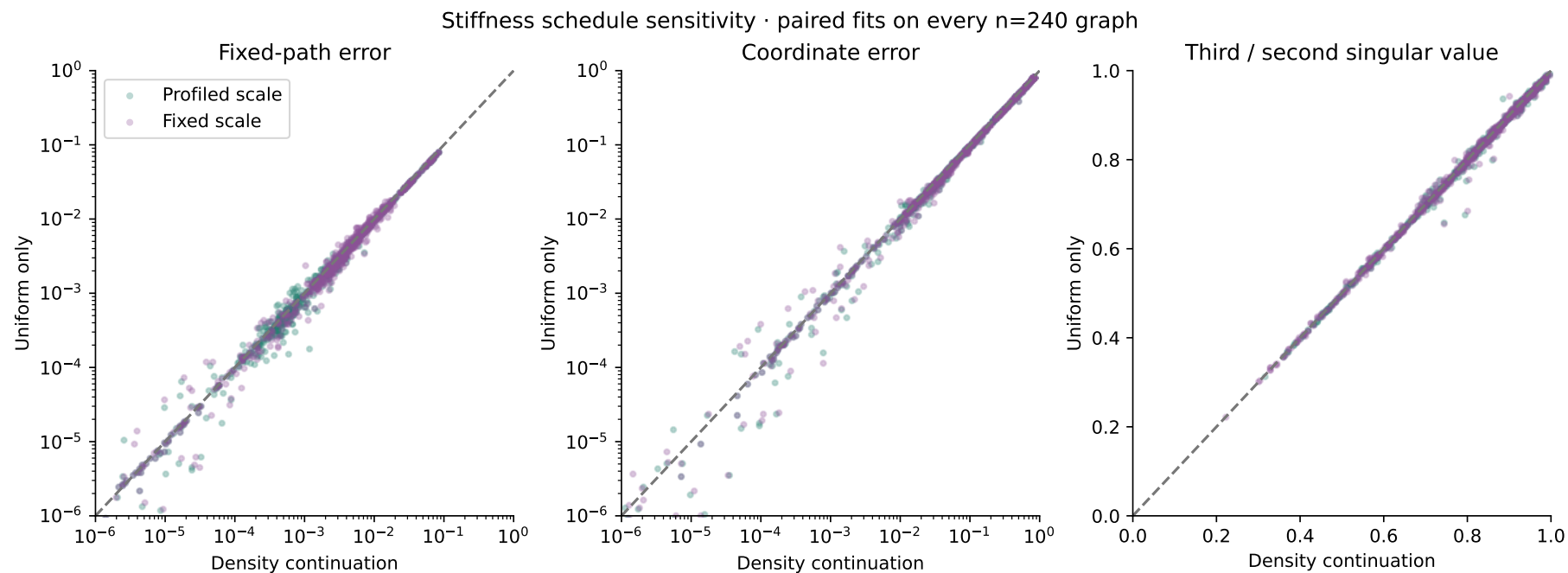
Solid curves initialize from graph distances; dashed curves reuse full-geodesic classical MDS across k at each radius and sample. Both use fixed-scale density continuation. Medians over three samples; both measures are shown. Panels use separate logarithmic y ranges. Changing k can still alter the refined geometry even when the starting configuration is fixed.

Sensitivity to starts and additional steps



Each point compares a stress-MDS-initialized primary fit with a separate perturbation, random/original start, or extra-budget fit. Both scale policies are shown. Logarithmic panels omit values below 10^{-6} ; all values remain in the tables. These diagnose particular fits, not global optimality.

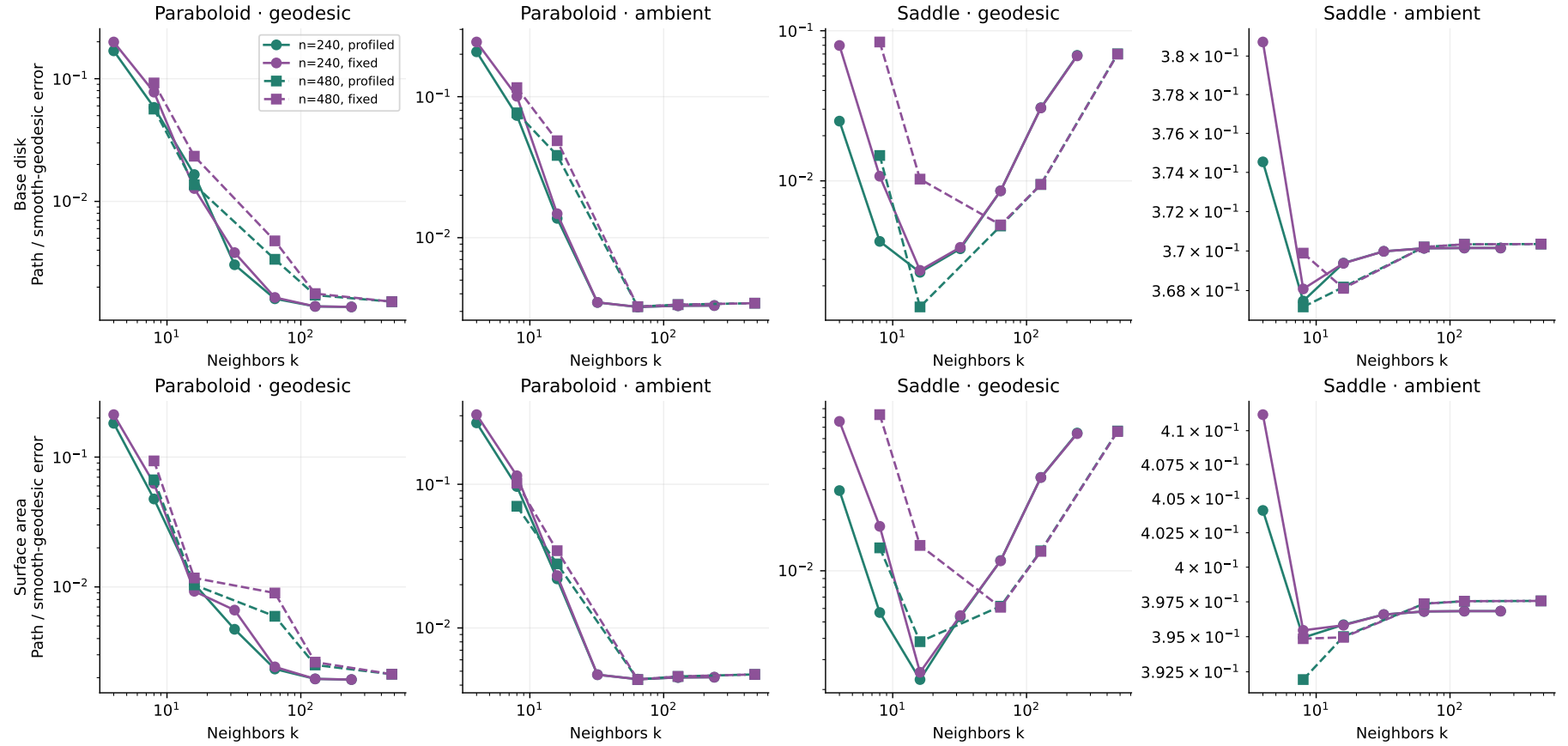
Uniform stiffness versus density continuation



Paired runs use the same 1,000-step allowance. All radii, k values, sampling measures and replicates are included. Values below 10^{-6} lie outside logarithmic panels. The diagonal denotes equal values, not an optimality condition.

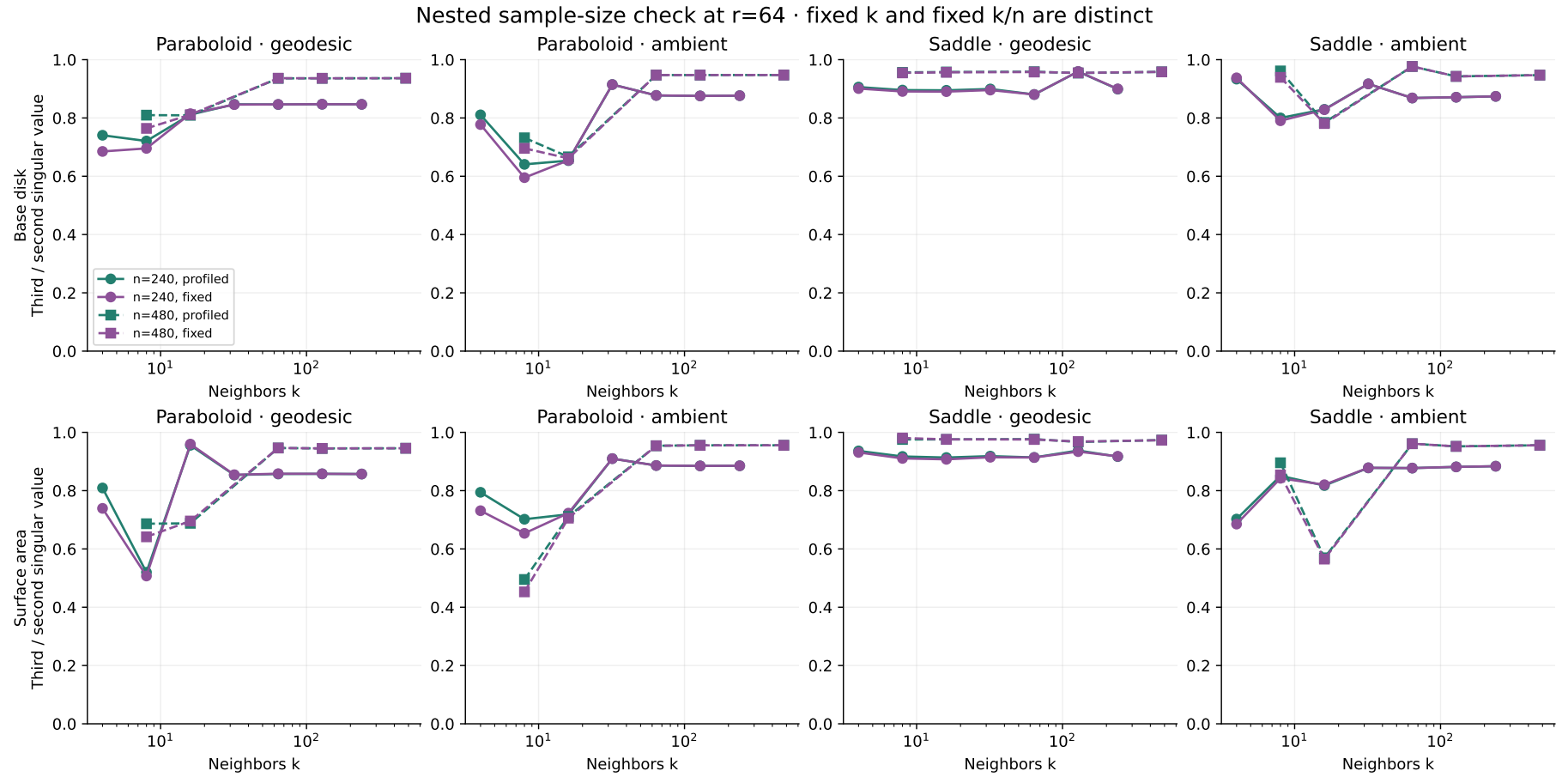
Nested sample-size sensitivity

Nested sample-size check at $r=64$ · fixed k and fixed k/n are distinct



At $r=64$, the $n=480$ sample retains the original 240 points. Solid and dashed lines identify the two sample sizes; colors identify scale policies. Fixed k and fixed k/n comparisons are distinct. Path-error panels use separate logarithmic y ranges; shape-ratio panels share $[0,1]$. Only one nested sample is available, so these are sensitivity checks, not large- n estimates.

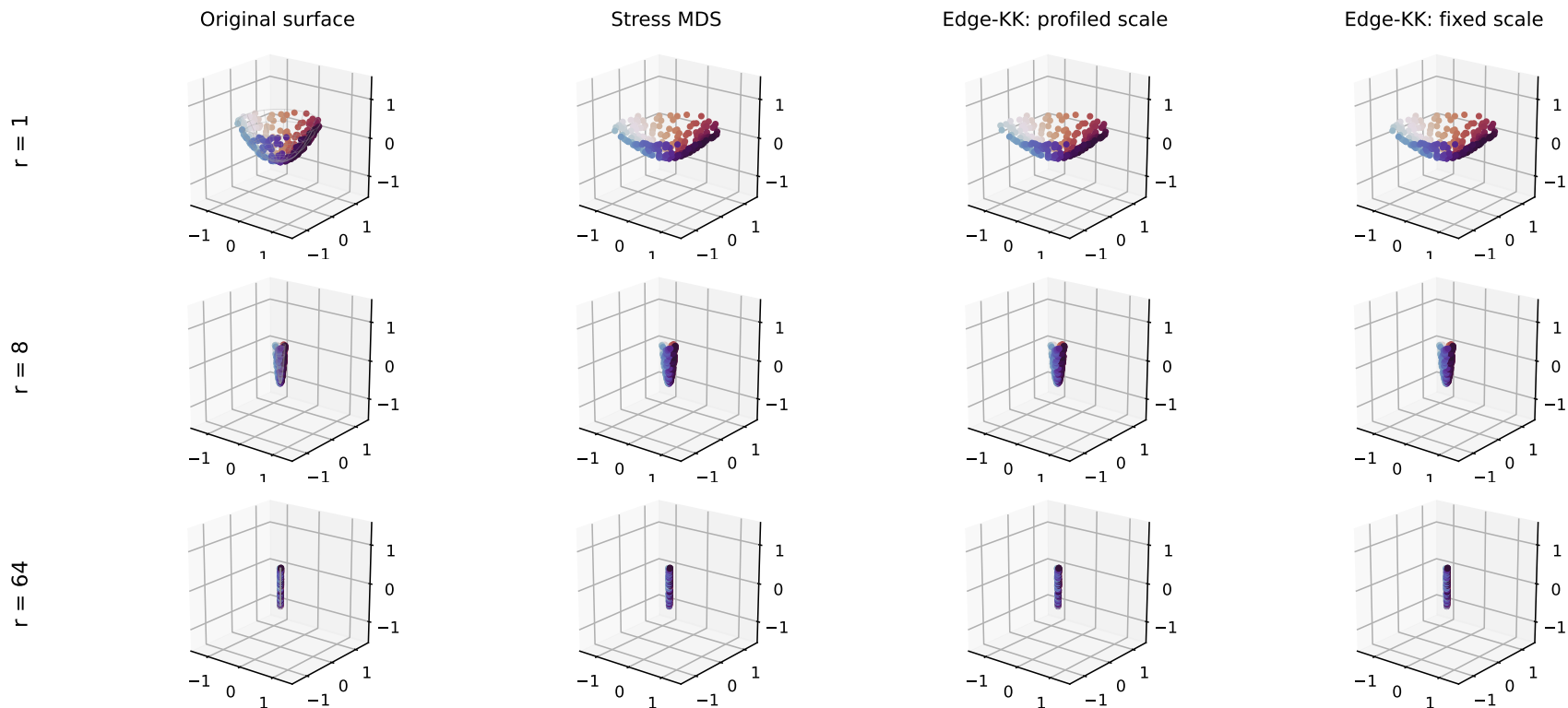
Nested sample-size sensitivity



At $r=64$, the $n=480$ sample retains the original 240 points. Solid and dashed lines identify the two sample sizes; colors identify scale policies. Fixed k and fixed k/n comparisons are distinct. Path-error panels use separate logarithmic y ranges; shape-ratio panels share $[0,1]$. Only one nested sample is available, so these are sensitivity checks, not large- n estimates.

Complete-graph spatial control: paraboloid, geodesic

Paraboloid · geodesic graph · $k=239$ · first base-disk sample
 Edge-calibrated physical coordinates / r^2 ; equal spatial units and common limits

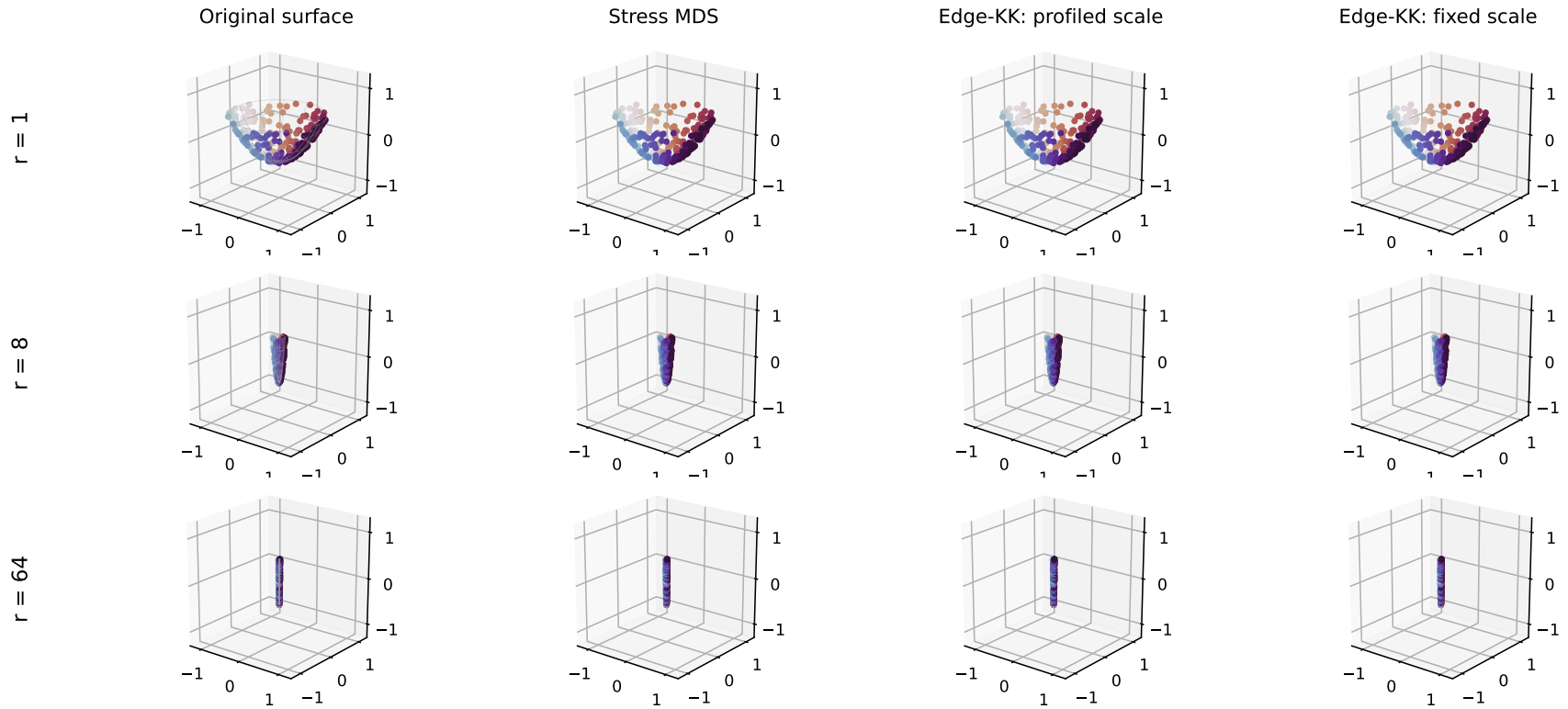


Original wireframe covers the full parameter disk. Thin equal-unit views do not mean the surface domain was truncated.

The complete ambient graph has an exact original-coordinate realization. The geodesic control tests global distance constraints. Coordinates use edge calibration followed by one uniform r^2 normalization, equal axis units, and common limits. First base-disk sample only.

Complete-graph spatial control: paraboloid, ambient

Paraboloid · ambient graph · $k=239$ · first base-disk sample
 Edge-calibrated physical coordinates / r^2 ; equal spatial units and common limits

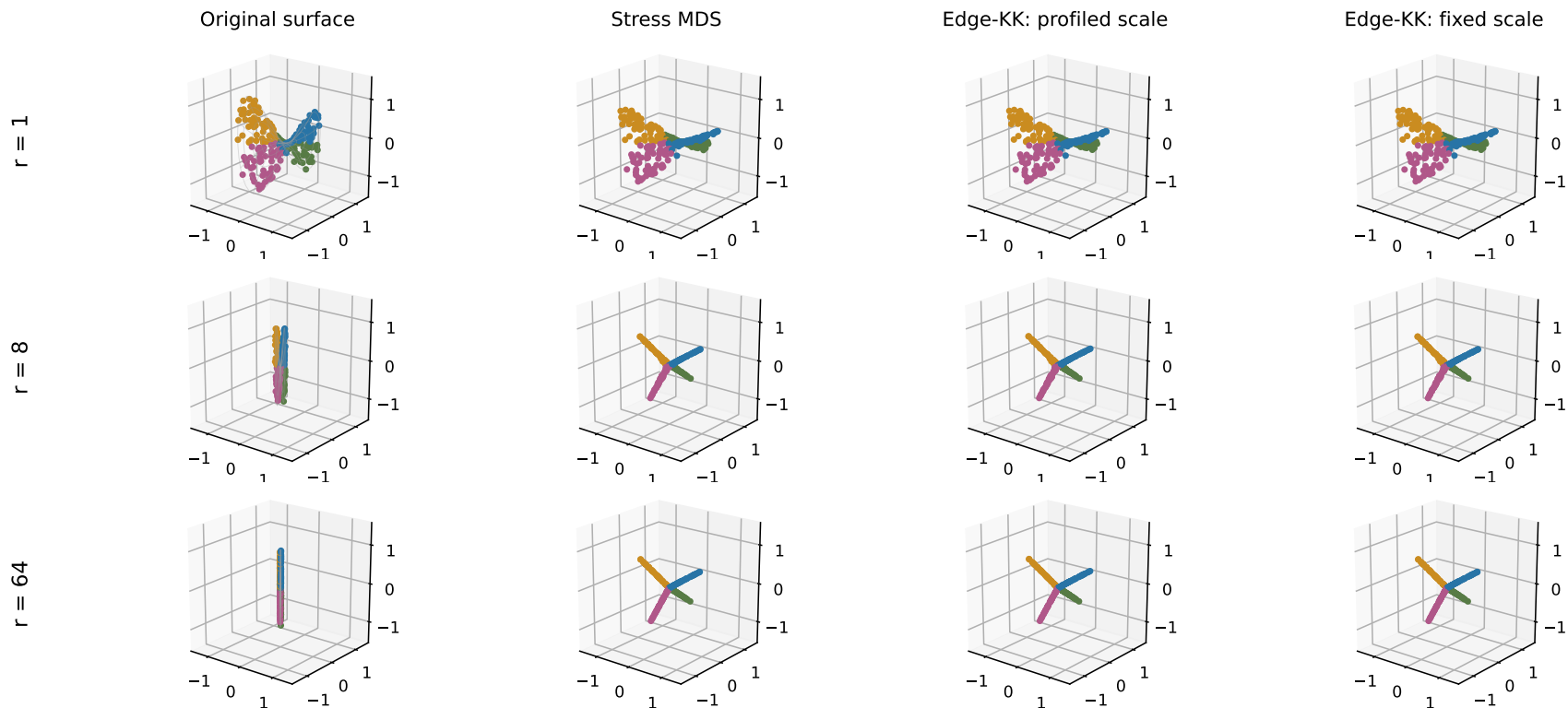


Original wireframe covers the full parameter disk. Thin equal-unit views do not mean the surface domain was truncated.

The complete ambient graph has an exact original-coordinate realization. The geodesic control tests global distance constraints. Coordinates use edge calibration followed by one uniform r^2 normalization, equal axis units, and common limits. First base-disk sample only.

Complete-graph spatial control: saddle, geodesic

Saddle · geodesic graph · $k=239$ · first base-disk sample
 Edge-calibrated physical coordinates / r^2 ; equal spatial units and common limits

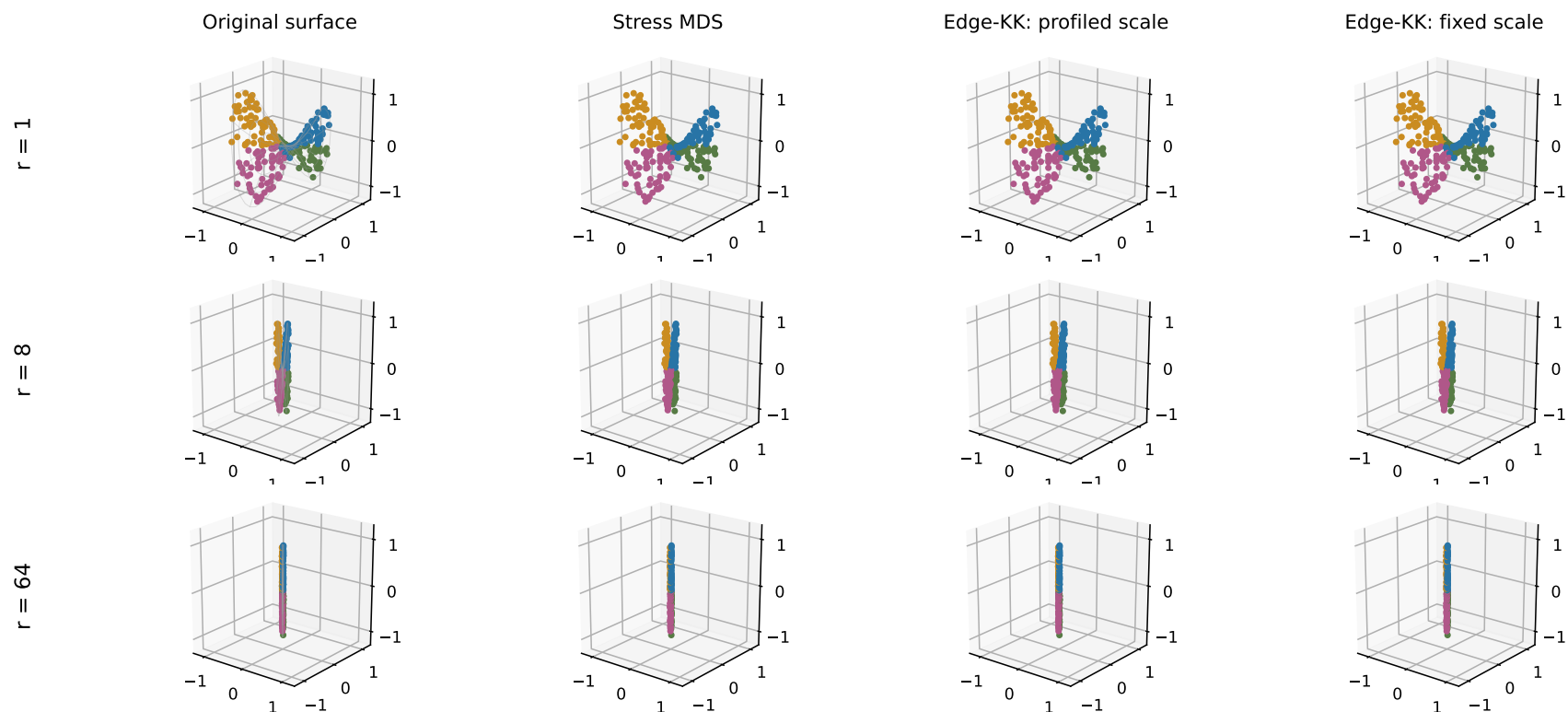


Original wireframe covers the full parameter disk. Thin equal-unit views do not mean the surface domain was truncated.

The complete ambient graph has an exact original-coordinate realization. The geodesic control tests global distance constraints. Coordinates use edge calibration followed by one uniform r^2 normalization, equal axis units, and common limits. First base-disk sample only.

Complete-graph spatial control: saddle, ambient

Saddle · ambient graph · $k=239$ · first base-disk sample
 Edge-calibrated physical coordinates / r^2 ; equal spatial units and common limits



Original wireframe covers the full parameter disk. Thin equal-unit views do not mean the surface domain was truncated.

The complete ambient graph has an exact original-coordinate realization. The geodesic control tests global distance constraints. Coordinates use edge calibration followed by one uniform r^2 normalization, equal axis units, and common limits. First base-disk sample only.

Reproducibility and references

The accompanying protocol, R/Python sources, compact distance matrices, coordinates, full stratified tables, input/source hashes, and validation records form the reproducibility bundle. Data-generation and fitting caches are separate from the compact artifacts used to render this report. No build or validation step requires private agent notes.

References

Patrick Mair, Patrick J. F. Groenen, and Jan de Leeuw. More on multidimensional scaling and unfolding in R: smacof version 2. *Journal of Statistical Software*, 102(10):1–47, 2022. doi: 10.18637/jss.v102.i10. URL <https://www.jstatsoft.org/article/view/v102i10>.